Mathematical Analysis II

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Introduction

The first volume of these notes develops the basic theory of one-variable mathematical analysis: limits, continuity, differentiation, and integration. The present volume continues from that foundation and studies what happens when the objects under consideration depend on several variables or occupy regions, curves, and surfaces in higher-dimensional space.

The transition from one variable to several variables is not merely a change of notation. For a function of one variable, an increment is described by a single number. For a function of several variables, different variables may change simultaneously, and the behavior of the function must be studied with respect to several directions. This leads naturally to partial derivatives, total differentials, directional derivatives, and the differential theory of multivariable functions.

Multivariable calculus grew together with the development of mechanics, geometry, and mathematical physics. The differential and integral calculus created by Newton and Leibniz provided the basic language for describing continuously varying quantities. During the eighteenth and nineteenth centuries, mathematicians including Euler, Lagrange, Gauss, Green, and Stokes extended these ideas to problems involving several variables, curves, surfaces, and spatial regions. Many formulas that first appeared in concrete geometric or physical problems gradually became fundamental theorems of analysis.

This volume begins with *multivariable differentiation*. The notions of limit and continuity developed in one variable are extended to functions of several variables, after which partial derivatives, differentials, derivatives of composite functions, implicit functions, and extrema are studied. The main purpose is to understand how the local behavior of a multivariable function can be described through its derivatives.

The second chapter develops *multiple integrals*. The definite integral of one variable is replaced by integration over two-dimensional and higher-dimensional regions. The definition, properties, computation, and change of variables of multiple integrals form the main subjects of this chapter.

The next two chapters concern integration along curves and over surfaces. Line integrals allow quantities to be accumulated along a curve, while surface integrals perform the analogous operation over a surface. These ideas lead to several of the central integral formulas of multivariable calculus and connect local differential information with global integral quantities.

Integration also has important geometric and quantitative applications. A separate chapter is therefore devoted to the applications of integration contained in the course. These problems show how the integral can be used not only as an analytic construction but also as a method for computing quantities arising from geometry and related applications.

The final chapter returns to the geometric structure underlying multivariable analysis. In $\mathbb{R}^n$, concepts such as neighborhoods, open and closed sets, convergence of point sequences, compactness, and connectedness provide the language needed to formulate continuity and other properties of multivariable functions precisely. Only the point-set concepts required in the original course are developed here; the purpose is to support multivariable analysis rather than to build a separate theory of general topology.

Thus the organization of this volume may be viewed schematically as

$$\text{multivariable differentiation} \longrightarrow \text{multiple integration} \longrightarrow \text{line and surface integration}$$

followed by the applications and geometric foundations of these ideas. Compared with

the first volume, the main change is that the basic operations of calculus are now carried out on higher-dimensional objects.

As in the first volume, the definitions, theorems, and principal proofs are kept close to the original course material and to *Mathematical Analysis* by Gengzhe Chang and Jihuai Shi. Computational arguments are shortened when appropriate, while the mathematical assumptions and conclusions of the original results are preserved.

The purpose of this volume is therefore not to extend the subject beyond the scope of the course, but to organize the multivariable part of mathematical analysis into a coherent continuation of the theory developed in the first volume.

Chapter 1

Multivariable Differentiation

In one-variable calculus, the local behavior of a function is described by its derivative. For functions of several variables, the situation is more complicated: the independent variables may change simultaneously and in different directions.
The purpose of this chapter is to extend the basic concepts of limits, continuity, derivatives, and differentials to functions of several variables.

1.1 Functions of Several Variables

Let D be a subset of $\mathbb{R}^n$.

Definition 1.1.1. If, for every point

$$P = (x_1, x_2, \ldots, x_n) \in D,$$

there corresponds a unique real number u, then u is called a function of n variables on D, written as

$$u = f(x_1, x_2, \ldots, x_n).$$

The set D is called the domain of the function.

For a function of two variables, we usually write

$$z = f(x, y).$$

Its domain is a subset of the xy-plane, and its graph is the set of points

$$(x, y, z)$$

satisfying

$$z = f(x, y).$$

For a fixed number c, the equation

$$f(x, y) = c$$

usually represents a curve in the xy-plane. Such a curve is called a level curve of f.
For a function of three variables,

$$u = f(x, y, z),$$

the equation

$$f(x, y, z) = c$$

usually represents a surface, called a level surface.

1.2 Limits of Multivariable Functions

Let

$$P = (x_1, \ldots, x_n), \qquad P_0 = (x_1^0, \ldots, x_n^0).$$

The distance between P and P_0 is

$$|P - P_0| = \sqrt{(x_1 - x_1^0)^2 + \cdots + (x_n - x_n^0)^2}.$$

For two variables,

$$P = (x, y), \qquad P_0 = (x_0, y_0),$$

and hence

$$|P - P_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}.$$

Definition 1.2.1. Let f be defined in a deleted neighborhood of P_0. If, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |P - P_0| < \delta$$

implies

$$|f(P) - A| < \varepsilon,$$

then A is called the limit of $f(P)$ as $P \to P_0$, written as

$$\lim_{P \to P_0} f(P) = A.$$

For a function of two variables, this is written as

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = A.$$

Unlike the one-variable case, the point P may approach P_0 along infinitely many different paths.
Therefore, if

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y)$$

exists, the same limit must be obtained along every path approaching (x_0, y_0).
Consequently, to prove that a limit does not exist, it is sufficient to find two paths approaching the same point along which the corresponding limits are different.

Example 1.2.2. Consider

$$f(x, y) = \frac{xy}{x^2 + y^2}, \qquad (x, y) \neq (0, 0).$$

Along $y = 0$,

$$f(x, 0) = 0.$$

Hence

$$\lim_{x\to 0} f(x, 0) = 0.$$

Along $y = x$,

$$f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Hence

$$\lim_{x\to 0} f(x, x) = \frac{1}{2}.$$

Since the two limits are different,

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}$$

does not exist.

The algebraic properties of limits remain valid. If

$$\lim_{P\to P_0} f(P) = A, \qquad \lim_{P\to P_0} g(P) = B,$$

then

$$\lim_{P\to P_0} \Big(f(P) + g(P)\Big) = A + B,$$

$$\lim_{P\to P_0} f(P)g(P) = AB,$$

and, if $B \neq 0$,

$$\lim_{P\to P_0} \frac{f(P)}{g(P)} = \frac{A}{B}.$$

1.3 Continuity of Multivariable Functions

Definition 1.3.1. Let f be defined in a neighborhood of P_0. The function f is called continuous at P_0 if

$$\lim_{P \to P_0} f(P) = f(P_0).$$

For a function of two variables,

$$z = f(x, y),$$

continuity at (x_0, y_0) means

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = f(x_0, y_0).$$

Equivalently, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

implies

$$|f(x, y) - f(x_0, y_0)| < \varepsilon.$$

Sums, products, and quotients of continuous functions are continuous where the corresponding operations are defined.
Composite functions of continuous functions are also continuous under the corresponding assumptions.

1.4 Partial Derivatives

For a function

$$z = f(x, y),$$

we may fix one independent variable and differentiate with respect to the other.

Definition 1.4.1. The partial derivative of f with respect to x at (x_0, y_0) is defined by

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the limit exists.

Similarly,

$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{k \to 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k}.$$

The partial derivatives are also denoted by

$$f_x, \qquad f_y,$$

or

$$D_x f, \qquad D_y f.$$

For a function of n variables,

$$u = f(x_1, \ldots, x_n),$$

the partial derivative with respect to x_i is

$$\frac{\partial f}{\partial x_i}.$$

Example 1.4.2. Let

$$f(x, y) = x^2 y + e^{xy}.$$

Then

$$f_x(x, y) = 2xy + ye^{xy},$$

and

$$f_y(x, y) = x^2 + xe^{xy}.$$

The existence of partial derivatives at a point does not by itself imply continuity at that point.

1.5 Total Differentials

Let

$$z = f(x, y).$$

When x and y have increments Δx and Δy, respectively, the corresponding increment of z is

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y).$$

Set

$$\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Definition 1.5.1. If

$$\Delta z = A\Delta x + B\Delta y + o(\rho)$$

as

$$\rho \to 0,$$

where A and B are independent of $\Delta x, \Delta y$, then f is called differentiable at (x, y).

The linear part

$$A\Delta x + B\Delta y$$

is called the total differential of f.

Theorem 1.5.2. If f is differentiable at (x, y), then its partial derivatives exist at that point, and

$$A = f_x(x, y), \qquad B = f_y(x, y).$$

Therefore

$$df = f_x\, dx + f_y\, dy.$$

For

$$z = f(x, y),$$

we also write

$$dz = \frac{\partial z}{\partial x}\, dx + \frac{\partial z}{\partial y}\, dy.$$

For a function of n variables,

$$u = f(x_1, \ldots, x_n),$$

the total differential is

$$du = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\, dx_i.$$

Theorem 1.5.3. If f is differentiable at P_0, then f is continuous at P_0.

Proof. By differentiability,

$$\Delta f = \sum_{i=1}^{n} A_i \Delta x_i + o(\rho),$$

where

$$\rho = \sqrt{(\Delta x_1)^2 + \cdots + (\Delta x_n)^2}.$$

Since

$$|\Delta x_i| \leq \rho,$$

we have

$$\sum_{i=1}^{n} A_i \Delta x_i \to 0$$

as $\rho \to 0$. Also,

$$o(\rho) \to 0.$$

Hence

$$\Delta f \to 0,$$

which proves continuity. □

Thus

$$\text{differentiability} \Longrightarrow \text{continuity}.$$

The converse is not generally true.
Moreover, the existence of all partial derivatives does not necessarily imply differentiability.

1.6 A Sufficient Condition for Differentiability

Theorem 1.6.1. Suppose the partial derivatives

$$f_x(x, y), \qquad f_y(x, y)$$

exist in a neighborhood of (x_0, y_0) and are continuous at (x_0, y_0). Then f is differentiable at (x_0, y_0).

Proof. Write

$$\begin{aligned}\Delta z &= f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \\ &= \Big[f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)\Big] \\ &\quad + \Big[f(x_0, y_0 + \Delta y) - f(x_0, y_0)\Big].\end{aligned}$$

Applying the one-variable mean value theorem to the two terms, there exist suitable intermediate points such that

$$\Delta z = f_x(\xi, y_0 + \Delta y)\Delta x + f_y(x_0, \eta)\Delta y.$$

By continuity of the partial derivatives,

$$f_x(\xi, y_0 + \Delta y) = f_x(x_0, y_0) + o(1)$$

and

$$f_y(x_0, \eta) = f_y(x_0, y_0) + o(1).$$

Therefore

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + o(\rho).$$

Hence f is differentiable at (x_0, y_0). □

1.7 Directional Derivatives

Partial derivatives describe changes parallel to the coordinate axes. More generally, one may study the change of a function in an arbitrary direction.
Let

$$\boldsymbol{l} = (l_1, \ldots, l_n)$$

be a unit vector.

Definition 1.7.1. The directional derivative of f at P in the direction $\boldsymbol{l}$ is

$$D_{\boldsymbol{l}} f(P) = \lim_{t \to 0} \frac{f(P + t\boldsymbol{l}) - f(P)}{t},$$

provided the limit exists.

For a function $f(x, y)$, if

$$\boldsymbol{l} = (\cos\alpha, \cos\beta),$$

then

$$D_{\boldsymbol{l}} f = f_x \cos\alpha + f_y \cos\beta,$$

provided f is differentiable.

Definition 1.7.2. For a differentiable function

$$f(x_1, \ldots, x_n),$$

the vector

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right)$$

is called the gradient of f.

Thus

$$D_{\boldsymbol{l}} f = \nabla f \cdot \boldsymbol{l}.$$

Since $\boldsymbol{l}$ is a unit vector,

$$D_{\boldsymbol{l}} f = \|\nabla f\| \cos\theta,$$

where θ is the angle between ∇f and $\boldsymbol{l}$.
Therefore the directional derivative has its maximum value

$$\|\nabla f\|$$

in the direction of ∇f.

1.8 Differentiation of Composite Functions

Suppose

$$z = f(u, v),$$

where

$$u = u(x, y), \qquad v = v(x, y).$$

Theorem 1.8.1 (Chain Rule)**.** Under the corresponding differentiability assumptions,

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x},$$

and

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y}.$$

In differential form,

$$dz = f_u\,du + f_v\,dv.$$

Since

$$du = u_x\,dx + u_y\,dy$$

and

$$dv = v_x\,dx + v_y\,dy,$$

we obtain

$$\begin{aligned} dz = f_u\Big(u_x\,dx + u_y\,dy\Big) \\ + f_v\Big(v_x\,dx + v_y\,dy\Big). \end{aligned}$$

If

$$z = f(u,v), \qquad u = u(t), \qquad v = v(t),$$

then

$$\frac{dz}{dt} = \frac{\partial f}{\partial u}\frac{du}{dt} + \frac{\partial f}{\partial v}\frac{dv}{dt}.$$

1.9 Higher-Order Partial Derivatives

If the first-order partial derivatives are themselves differentiable, we may differentiate them again.
For a function $f(x,y)$, the second-order partial derivatives are

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{xy} = \frac{\partial^2 f}{\partial y \partial x},$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}.$$

The derivatives

$$f_{xy}, \qquad f_{yx}$$

are called mixed partial derivatives.

Theorem 1.9.1. If the mixed partial derivatives

$$f_{xy} \qquad \text{and} \qquad f_{yx}$$

are continuous in a neighborhood of a point, then

$$f_{xy} = f_{yx}$$

at that point.

Thus, under suitable continuity assumptions, the order of differentiation may be interchanged.

1.10 Higher-Order Differentials

For

$$z = f(x, y),$$

the first differential is

$$dz = f_x\,dx + f_y\,dy.$$

Regarding dx and dy as independent of x and y, the second differential is

$$d^2z = d(dz).$$

Hence

$$d^2z = f_{xx}\,dx^2 + 2f_{xy}\,dx\,dy + f_{yy}\,dy^2.$$

Symbolically,

$$d^2z = \left(dx\frac{\partial}{\partial x} + dy\frac{\partial}{\partial y}\right)^2 f.$$

More generally,

$$d^nz = \left(dx\frac{\partial}{\partial x} + dy\frac{\partial}{\partial y}\right)^n f,$$

provided the required derivatives exist and the corresponding mixed derivatives can be interchanged.

1.11 Taylor Formula for Functions of Several Variables

Taylor's formula extends the local polynomial approximation from one variable to several variables.

For a function $f(x, y)$, let

$$x = x_0 + h, \qquad y = y_0 + k.$$

The first-order approximation is

$$\begin{aligned} f(x_0 + h, y_0 + k) = {} & f(x_0, y_0) \\ & + f_x(x_0, y_0)h + f_y(x_0, y_0)k + o(\rho), \end{aligned}$$

where

$$\rho = \sqrt{h^2 + k^2}.$$

If the required second derivatives exist, the second-order Taylor formula takes the form

$$\begin{aligned} f(x_0 + h, y_0 + k) = {} & f(x_0, y_0) + f_x(x_0, y_0)h \\ & + f_y(x_0, y_0)k + \frac{1}{2!}\big[f_{xx}(x_0, y_0)h^2 \\ & \quad + 2f_{xy}(x_0, y_0)hk + f_{yy}(x_0, y_0)k^2\big] \\ & + o(\rho^2). \end{aligned}$$

Using differential notation, this may be written as

$$f(P + \Delta P) = f(P) + df + \frac{1}{2!}d^2f + o(\rho^2).$$

More generally, the Taylor formula can be expressed through successive differentials.

1.12 Implicit Functions

Consider an equation

$$F(x, y) = 0.$$

Under suitable conditions, this equation may determine y as a function of x near a given point.

Theorem 1.12.1 (Implicit Function Theorem)**.** Suppose

$$F(x_0, y_0) = 0,$$

and the partial derivatives F_x and F_y are continuous in a neighborhood of (x_0, y_0). If

$$F_y(x_0, y_0) \neq 0,$$

then, in a sufficiently small neighborhood of x_0, the equation

$$F(x, y) = 0$$

uniquely determines a continuous function

$$y = f(x)$$

satisfying

$$f(x_0) = y_0.$$

Moreover, f is differentiable and

$$\boxed{\frac{dy}{dx} = -\frac{F_x}{F_y}.}$$

Indeed, differentiating

$$F(x, y(x)) = 0$$

with respect to x gives

$$F_x + F_y \frac{dy}{dx} = 0.$$

Therefore

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

For a system

$$\begin{cases} F(x, y, u, v) = 0, \\ G(x, y, u, v) = 0, \end{cases}$$

the possibility of solving for u, v as functions of x, y is related to the determinant

$$\frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}.$$

This determinant is a Jacobian determinant.

1.13 Jacobians

Let

$$u = u(x, y), \qquad v = v(x, y).$$

Definition 1.13.1. The determinant

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}$$

is called the Jacobian determinant of u, v with respect to x, y.

Thus

$$\frac{\partial(u, v)}{\partial(x, y)} = u_x v_y - u_y v_x.$$

For n functions

$$y_i = y_i(x_1, \ldots, x_n), \qquad i = 1, \ldots, n,$$

the Jacobian is

$$\frac{\partial(y_1, \ldots, y_n)}{\partial(x_1, \ldots, x_n)} = \det\left(\frac{\partial y_i}{\partial x_j}\right).$$

Jacobians will also appear later in the change of variables for multiple integrals.

1.14 Extrema of Functions of Several Variables

Definition 1.14.1. Let f be defined in a neighborhood of P_0. If there exists a neighborhood of P_0 such that

$$f(P) \le f(P_0)$$

for every P in that neighborhood, then $f(P_0)$ is called a local maximum.
If

$$f(P) \ge f(P_0),$$

then $f(P_0)$ is called a local minimum.

Local maxima and local minima are collectively called local extrema.

Theorem 1.14.2. Suppose $f(x, y)$ has a local extremum at an interior point (x_0, y_0), and its partial derivatives exist there. Then

$$f_x(x_0, y_0) = 0, \qquad f_y(x_0, y_0) = 0.$$

Proof. Fix $y = y_0$. Then the one-variable function

$$\varphi(x) = f(x, y_0)$$

has a local extremum at x_0. By Fermat's theorem,

$$\varphi'(x_0) = 0.$$

Hence

$$f_x(x_0, y_0) = 0.$$

Similarly, fixing $x = x_0$ gives

$$f_y(x_0, y_0) = 0.$$

□

A point satisfying

$$f_x = f_y = 0$$

is called a stationary point.
The vanishing of the first partial derivatives is only a necessary condition and is not sufficient for a local extremum.

1.15 Second-Order Test for Extrema

Suppose

$$f_x(x_0, y_0) = f_y(x_0, y_0) = 0.$$

Set

$$A = f_{xx}(x_0, y_0), \qquad B = f_{xy}(x_0, y_0), \qquad C = f_{yy}(x_0, y_0).$$

The quadratic part of the Taylor expansion is

$$Ah^2 + 2Bhk + Ck^2.$$

Define

$$D = AC - B^2.$$

Theorem 1.15.1. Suppose the second partial derivatives are continuous near (x_0, y_0). Then:

1. if $D > 0$ and $A > 0$, then $f(x_0, y_0)$ is a local minimum;
2. if $D > 0$ and $A < 0$, then $f(x_0, y_0)$ is a local maximum;
3. if $D < 0$, then $f(x_0, y_0)$ is not a local extremum;
4. if $D = 0$, the test gives no conclusion.

1.16 Conditional Extrema

We now consider extrema subject to a constraint.
Suppose

$$z = f(x, y)$$

is to be extremized under the condition

$$\varphi(x, y) = 0.$$

Theorem 1.16.1 (Lagrange Multiplier Method)**.** At a conditional extremum satisfying the corresponding regularity conditions, there exists a number λ such that

$$\nabla f = \lambda \nabla \varphi.$$

Thus the candidate points are obtained from

$$\begin{cases} f_x = \lambda \varphi_x, \\ f_y = \lambda \varphi_y, \\ \varphi(x, y) = 0. \end{cases}$$

Equivalently, define the Lagrange function

$$L(x, y, \lambda) = f(x, y) - \lambda \varphi(x, y).$$

Then solve

$$L_x = 0, \qquad L_y = 0, \qquad L_\lambda = 0.$$

For a function of n variables subject to several constraints, the same principle introduces one multiplier for each constraint.

Chapter 2

Multiple Integrals

The definite integral of one variable is defined through limits of Riemann sums. The same idea extends naturally to functions defined on subsets of $\mathbb{R}^2$, $\mathbb{R}^3$, and, more generally, $\mathbb{R}^n$. We first develop the integral over a rectangle, where the construction is closest to the one-variable Riemann integral. We then pass to more general bounded sets, establish formulas for repeated integration and change of variables, and finally extend the theory to triple and n-fold integrals.

2.1 Integrals over Rectangles

Let

$$I = [a, b] \times [c, d]$$

be a rectangle in $\mathbb{R}^2$.
Take partitions

$$a = x_0 < x_1 < \cdots < x_n = b$$

and

$$c = y_0 < y_1 < \cdots < y_m = d.$$

These partitions divide I into subrectangles

$$I_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j].$$

For each I_{ij}, choose a point

$$\xi_{ij} \in I_{ij}.$$

The area of I_{ij} is

$$\sigma(I_{ij}) = \Delta x_i \Delta y_j,$$

where

$$\Delta x_i = x_i - x_{i-1}, \qquad \Delta y_j = y_j - y_{j-1}.$$

The corresponding Riemann sum is

$$\sum_{i=1}^{n} \sum_{j=1}^{m} f(\xi_{ij}) \sigma(I_{ij}).$$

Let

$$\|\pi\| = \max_{i,j} \operatorname{diam}(I_{ij}).$$

Definition 2.1.1. Let f be bounded on I. If there exists a real number A such that, for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever

$$\|\pi\| < \delta,$$

we have

$$\left| \sum_{i=1}^{n} \sum_{j=1}^{m} f(\xi_{ij}) \sigma(I_{ij}) - A \right| < \varepsilon,$$

then f is called Riemann integrable on I.

The number A is called the double integral of f over I, written as

$$\iint_I f(x, y)\, dx\, dy,$$

or more briefly,

$$\int_I f\, d\sigma.$$

If $f \equiv c$, then

$$\iint_I f\, d\sigma = c\, \sigma(I).$$

2.2 Basic Properties of Double Integrals

Suppose f and g are integrable on I.
For any constant c,

$$\int_I cf\, d\sigma = c\int_I f\, d\sigma.$$

Also,

$$\int_I (f+g)\, d\sigma = \int_I f\, d\sigma + \int_I g\, d\sigma.$$

If

$$f \geq 0$$

on I, then

$$\int_I f\, d\sigma \geq 0.$$

If

$$f \geq g,$$

then

$$\int_I f\, d\sigma \geq \int_I g\, d\sigma.$$

2.3 Upper and Lower Sums

Let

$$m_{ij} = \inf_{I_{ij}} f, \qquad M_{ij} = \sup_{I_{ij}} f.$$

The lower sum corresponding to the partition π is

$$S(f,\pi) = \sum_{i,j} m_{ij}\sigma(I_{ij}),$$

and the upper sum is

$$\overline{S}(f,\pi) = \sum_{i,j} M_{ij}\sigma(I_{ij}).$$

Clearly,

$$S(f,\pi) \leq \overline{S}(f,\pi).$$

If π' is a refinement of π, then

$$S(f,\pi) \leq S(f,\pi')$$

and

$$\overline{S}(f,\pi') \leq \overline{S}(f,\pi).$$

Definition 2.3.1. The lower integral is defined by

$$\underline{\int_I} f\, d\sigma = \sup_{\pi} S(f,\pi),$$

and the upper integral is defined by

$$\overline{\int_I} f\, d\sigma = \inf_{\pi} \overline{S}(f,\pi).$$

For every partition,

$$S(f,\pi) \leq \underline{\int_I} f\, d\sigma \leq \overline{\int_I} f\, d\sigma \leq \overline{S}(f,\pi).$$

Theorem 2.3.2. A bounded function f on I is Riemann integrable if and only if, for every $\varepsilon > 0$, there exists a partition π such that

$$\overline{S}(f, \pi) - S(f, \pi) < \varepsilon.$$

Equivalently,

$$\underline{\int_I} f \, d\sigma = \overline{\int_I} f \, d\sigma.$$

2.4 Oscillation and Integrability

For a subrectangle I_{ij}, define the oscillation of f by

$$\omega(f, I_{ij}) = M_{ij} - m_{ij}.$$

Then

$$\overline{S}(f, \pi) - S(f, \pi) = \sum_{i,j} \omega(f, I_{ij}) \sigma(I_{ij}).$$

Hence the integrability criterion may also be written as follows.

Theorem 2.4.1. A bounded function f on I is Riemann integrable if and only if, for every $\varepsilon > 0$, there exists a partition π such that

$$\sum_{i,j} \omega(f, I_{ij}) \sigma(I_{ij}) < \varepsilon.$$

In particular, every continuous function on a rectangle is Riemann integrable.

2.5 Zero-Measure Sets

The integrability criterion can be described in terms of the set of discontinuities.

Definition 2.5.1. A set $B \subset \mathbb{R}^2$ is called a two-dimensional zero-measure set if for every $\varepsilon > 0$, there exist finitely many rectangles

$$I_1, \ldots, I_m$$

such that

$$B \subset \bigcup_{k=1}^{m} I_k$$

and

$$\sum_{k=1}^{m} \sigma(I_k) < \varepsilon.$$

A finite set is a zero-measure set.
A countable set is also a zero-measure set.
The graph of a continuously differentiable function on a closed interval is a zero-measure set in $\mathbb{R}^2$.

2.6 Lebesgue Criterion

For a bounded function f, denote its set of discontinuities by

$$D(f).$$

Theorem 2.6.1 (Lebesgue Theorem)**.** Let f be bounded on a rectangle I. Then f is Riemann integrable on I if and only if

$$D(f)$$

is a two-dimensional zero-measure set.

Thus a bounded function with a zero-measure set of discontinuities is Riemann integrable.

Corollary 2.6.2. If f and g are Riemann integrable on I, and the set

$$\{p \in I : f(p) \neq g(p)\}$$

is a zero-measure set, then

$$\int_I f \, d\sigma = \int_I g \, d\sigma.$$

2.7 Repeated Integrals over Rectangles

Let

$$I = [a, b] \times [c, d].$$

Suppose f is integrable on I.
For fixed $x \in [a, b]$, consider the function

$$y \longmapsto f(x, y).$$

If it is integrable on $[c, d]$, define

$$\varphi(x) = \int_c^d f(x, y) \, dy.$$

Similarly, if for fixed y,

$$x \longmapsto f(x, y)$$

is integrable on $[a, b]$, define

$$\psi(y) = \int_a^b f(x, y) \, dx.$$

Theorem 2.7.1. If f is integrable on I, and φ is integrable on $[a, b]$, then

$$\boxed{\iint_I f(x, y) \, dx \, dy = \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx.}$$

Similarly,

$$\boxed{\iint_I f(x, y) \, dx \, dy = \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy.}$$

Theorem 2.7.2. If f is continuous on the rectangle I, then

$$\boxed{\iint_I f(x, y) \, dx \, dy = \int_a^b \int_c^d f(x, y) \, dy \, dx}$$

and

$$\boxed{\iint_I f(x, y) \, dx \, dy = \int_c^d \int_a^b f(x, y) \, dx \, dy.}$$

Thus, for continuous functions on a rectangle, the order of integration may be interchanged.

2.8 Integrals over Bounded Sets

Let $B \subset \mathbb{R}^2$ be bounded, and let

$$f : B \to \mathbb{R}.$$

Choose a rectangle I containing B. Define the extension f_I on I by

$$f_I(p) = \begin{cases} f(p), & p \in B, \\ 0, & p \notin B. \end{cases}$$

Definition 2.8.1. If f_I is Riemann integrable on I, then f is called integrable on B, and

$$\int_B f \, d\sigma = \int_I f_I \, d\sigma.$$

This definition does not depend on the particular rectangle I chosen to contain B.
For constants c,

$$\int_B cf \, d\sigma = c \int_B f \, d\sigma,$$

and

$$\int_B (f + g) \, d\sigma = \int_B f \, d\sigma + \int_B g \, d\sigma.$$

If

$$f \leq g$$

on B, then

$$\int_B f \, d\sigma \leq \int_B g \, d\sigma.$$

2.9 Area of a Bounded Set

Let

$$\chi_B(p) = \begin{cases} 1, & p \in B, \\ 0, & p \notin B. \end{cases}$$

Definition 2.9.1. If χ_B is integrable, then B is called measurable in the sense used here, and its area is defined by

$$\boxed{\sigma(B) = \int_B 1 \, d\sigma.}$$

Equivalently,

$$\sigma(B) = \int_I \chi_B \, d\sigma,$$

where I is any rectangle containing B.

Theorem 2.9.2. A bounded set $B \subset \mathbb{R}^2$ has area if and only if its boundary

$$\partial B$$

is a zero-measure set.

Thus a bounded region whose boundary consists of finitely many continuously differentiable curves has an area.
If

$$B_1 \cap B_2$$

has zero measure, then

$$\sigma(B_1 \cup B_2) = \sigma(B_1) + \sigma(B_2).$$

2.10 Riemann Sums over General Bounded Sets

Suppose $B \subset \mathbb{R}^2$ has area and f is integrable on B.
Let

$$T = \{D_1, \ldots, D_m\}$$

be a partition of B into sets having area, and choose

$$\xi_i \in D_i.$$

Define

$$S(f, T) = \sum_{i=1}^{m} f(\xi_i)\sigma(D_i).$$

Let

$$\|T\| = \max_i \operatorname{diam}(D_i).$$

Theorem 2.10.1. If f is integrable on B, then

$$\boxed{\lim_{\|T\|\to 0} \sum_{i=1}^{m} f(\xi_i)\sigma(D_i) = \int_B f\, d\sigma.}$$

2.11 Mean Value Theorem for Double Integrals

Theorem 2.11.1. Let $B \subset \mathbb{R}^2$ be a bounded connected region with area. Suppose f is continuous on B. Then there exists $\xi \in B$ such that

$$\boxed{\int_B f\, d\sigma = f(\xi)\sigma(B).}$$

More generally, if g is integrable and

$$g \geq 0,$$

then under the corresponding assumptions there exists $\xi \in B$ such that

$$\int_B fg\, d\sigma = f(\xi) \int_B g\, d\sigma.$$

2.12 Computation over Bounded Regions

Suppose

$$B = \{(x, y) : a \leq x \leq b,\ y_1(x) \leq y \leq y_2(x)\}.$$

If the corresponding functions satisfy the assumptions in the textbook, then

$$\boxed{\int_B f\, d\sigma = \int_a^b \left[\int_{y_1(x)}^{y_2(x)} f(x, y)\, dy \right] dx.}$$

Similarly, if

$$B = \{(x, y) : c \leq y \leq d,\ x_1(y) \leq x \leq x_2(y)\},$$

then

$$\boxed{\int_B f\, d\sigma = \int_c^d \left[\int_{x_1(y)}^{x_2(y)} f(x, y)\, dx \right] dy.}$$

Thus an important part of evaluating a double integral is correctly describing the integration region.

2.13 Change of Variables in Double Integrals

Let

$$x = x(u, v), \qquad y = y(u, v)$$

define a one-to-one continuously differentiable mapping from a region Δ in the uv-plane onto a region D in the xy-plane.
The Jacobian determinant is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix}.$$

Theorem 2.13.1 (Change of Variables Formula)**.** Under the assumptions of the change of variables theorem,

$$\boxed{\iint_D F(x, y)\, dx\, dy = \iint_\Delta F(x(u, v), y(u, v)) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv.}$$

The absolute value of the Jacobian appears because the Jacobian describes the local change of area under the mapping.

2.14 Polar Coordinates

The most frequently used change of variables in the plane is

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Its Jacobian is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r.$$

Therefore

$$\boxed{dx\, dy = r\, dr\, d\theta.}$$

Hence

$$\iint_D f(x, y)\, dx\, dy = \iint_\Delta f(r\cos\theta, r\sin\theta) r\, dr\, d\theta.$$

Polar coordinates are especially useful when the domain or the integrand contains expressions involving

$$x^2 + y^2.$$

2.15 Triple Integrals

The construction of a triple integral is analogous to that of a double integral.
Let $V \subset \mathbb{R}^3$ be a bounded region, and divide V into small pieces

$$V_1, \ldots, V_m.$$

Choose

$$P_i \in V_i.$$

The corresponding Riemann sum is

$$\sum_{i=1}^{m} f(P_i)\mu(V_i),$$

where $\mu(V_i)$ denotes the volume of V_i.

Definition 2.15.1. If the limit

$$\lim_{\|T\|\to 0} \sum_{i=1}^{m} f(P_i)\mu(V_i)$$

exists and is independent of the partition and the sample points, then f is integrable on V, and the limit is called the triple integral:

$$\iiint_V f(x,y,z)\,dx\,dy\,dz.$$

When $f \equiv 1$,

$$\mu(V) = \iiint_V 1\,dx\,dy\,dz$$

is the volume of V.
The corresponding concepts of upper and lower sums, zero-measure sets, and the Lebesgue criterion are analogous to the two-dimensional case.

2.16 Computation of Triple Integrals

Suppose

$$V = \{(x,y,z) : (x,y) \in D,\ \varphi_1(x,y) \le z \le \varphi_2(x,y)\}.$$

Then

$$\boxed{\iiint_V f(x,y,z)\,dx\,dy\,dz = \iint_D \left[\int_{\varphi_1(x,y)}^{\varphi_2(x,y)} f(x,y,z)\,dz\right] dx\,dy.}$$

By describing the same region in another order, one may obtain other repeated-integral formulas.

2.17 Change of Variables in Triple Integrals

Suppose

$$x = x(u,v,w), \qquad y = y(u,v,w), \qquad z = z(u,v,w)$$

defines the transformation.
Its Jacobian is

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \det \begin{pmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{pmatrix}.$$

Under the assumptions of the change of variables theorem,

$$\boxed{\iiint_D F(x,y,z)\,dx\,dy\,dz = \iiint_\Delta F(x(u,v,w),y(u,v,w),z(u,v,w)) \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\,dv\,dw.}$$

2.18 Spherical Coordinates

The spherical coordinate transformation used in the textbook is

$$x = r\sin\theta\cos\varphi,$$

$$y = r\sin\theta\sin\varphi,$$

and

$$z = r\cos\theta.$$

Its Jacobian is

$$\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = r^2 \sin\theta.$$

Hence

$$\boxed{dx\,dy\,dz = r^2 \sin\theta\,dr\,d\theta\,d\varphi.}$$

This transformation is useful for regions bounded by spheres or cones.

2.19 n-Fold Integrals

The same construction extends to $\mathbb{R}^n$.
Let

$$I = I_1 \times \cdots \times I_n,$$

where

$$I_i = [a_i, b_i].$$

The n-dimensional volume of I is

$$\mu_n(I) = \prod_{i=1}^{n} (b_i - a_i).$$

After dividing I into small n-dimensional rectangles and forming the corresponding Riemann sums, one obtains the n-fold integral

$$\int_I f(x_1, \ldots, x_n)\,dx_1 \cdots dx_n.$$

The integral may also be written as

$$\int_I f\,d\mu_n.$$

The theories of integrability, repeated integration, zero-measure sets, and change of variables extend in the same way.

2.20 Repeated Integration in Higher Dimensions

For a suitable region $V \subset \mathbb{R}^n$, an n-fold integral can be reduced to repeated integrals. For example, if

$$V = \{(x_1, \ldots, x_n) : (x_1, \ldots, x_{n-1}) \in D,\ a(x_1, \ldots, x_{n-1}) \le x_n \le b(x_1, \ldots, x_{n-1})\},$$

then

$$\int_V f\,d\mu_n = \int_D \left[\int_{a(x_1,\ldots,x_{n-1})}^{b(x_1,\ldots,x_{n-1})} f(x_1, \ldots, x_n)\,dx_n \right] d\mu_{n-1}.$$

Thus the computation of multiple integrals ultimately reduces to a sequence of one-variable integrations when the region can be described appropriately.

2.21 Change of Variables in Higher Dimensions

Let

$$x_i = x_i(u_1, \ldots, u_n), \qquad i = 1, \ldots, n.$$

The Jacobian determinant is

$$\frac{\partial(x_1, \ldots, x_n)}{\partial(u_1, \ldots, u_n)} = \det\left(\frac{\partial x_i}{\partial u_j}\right).$$

Under the assumptions of the change of variables theorem,

$$\boxed{\int_D F(x_1, \ldots, x_n)\, dx_1 \cdots dx_n = \int_\Delta F(x(u)) \left|\frac{\partial(x_1, \ldots, x_n)}{\partial(u_1, \ldots, u_n)}\right| du_1 \cdots du_n.}$$

Chapter 3

Line Integrals

Multiple integrals integrate a function over a region. We now consider integration over a curve.
There are two types of line integrals. The first type is associated with quantities distributed along a curve and does not depend on the orientation of the curve. The second type arises naturally from work done by a vector field and depends essentially on the orientation of the curve.

3.1 Line Integrals of the First Type

Let Γ be a rectifiable curve in $\mathbb{R}^3$. Suppose a mass is distributed along Γ with linear density

$$\rho = \rho(x, y, z).$$

Divide Γ into small arcs

$$\Delta s_1, \Delta s_2, \ldots, \Delta s_n,$$

and choose a point

$$\xi_i \in \Delta s_i.$$

The mass of the i-th small arc is approximately

$$\rho(\xi_i)\Delta s_i.$$

Hence the total mass is approximated by

$$\sum_{i=1}^{n} \rho(\xi_i)\Delta s_i.$$

This motivates the following definition.

Definition 3.1.1. Let Γ be a rectifiable curve and let f be defined on Γ. If the limit

$$\lim_{\max \Delta s_i \to 0} \sum_{i=1}^{n} f(\xi_i)\Delta s_i$$

exists and is independent of the division and the choice of ξ_i, then this limit is called the line integral of the first type of f along Γ, denoted by

$$\int_\Gamma f\, ds.$$

For a mass density ρ, the total mass is therefore

$$m = \int_\Gamma \rho\, ds.$$

If $f \equiv 1$, then

$$\int_\Gamma ds = s(\Gamma),$$

where $s(\Gamma)$ is the length of Γ.
The first-type line integral has the usual linearity properties:

$$\int_\Gamma \big(af + bg\big)\, ds = a\int_\Gamma f\, ds + b\int_\Gamma g\, ds.$$

If $f \geq 0$ on Γ, then

$$\int_\Gamma f\, ds \geq 0.$$

If $f \leq g$, then

$$\int_\Gamma f\, ds \leq \int_\Gamma g\, ds.$$

3.2 Computation of First-Type Line Integrals

Suppose Γ has the parametrization

$$\boldsymbol{r} = \boldsymbol{r}(t), \qquad a \le t \le b,$$

where

$$\boldsymbol{r}(t) = \Big(x(t), y(t), z(t)\Big).$$

Then

$$ds = \|\boldsymbol{r}'(t)\|\, dt.$$

Theorem 3.2.1. Under the corresponding assumptions,

$$\boxed{\int_\Gamma f\, ds = \int_a^b f(\boldsymbol{r}(t))\, \|\boldsymbol{r}'(t)\|\, dt.}$$

In coordinates,

$$\boxed{\int_\Gamma f(x,y,z)\, ds = \int_a^b f(x(t),y(t),z(t))\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\, dt.}$$

For a plane curve

$$\boldsymbol{r}(t) = \Big(x(t), y(t)\Big),$$

the formula becomes

$$\int_\Gamma f(x,y)\, ds = \int_a^b f(x(t),y(t))\sqrt{(x'(t))^2 + (y'(t))^2}\, dt.$$

If the plane curve is given by

$$y = \varphi(x), \qquad a \le x \le b,$$

then

$$ds = \sqrt{1 + (\varphi'(x))^2}\, dx,$$

and therefore

$$\boxed{\int_\Gamma f(x,y)\, ds = \int_a^b f(x,\varphi(x))\sqrt{1 + (\varphi'(x))^2}\, dx.}$$

3.3 Orientation and First-Type Line Integrals

The value of a first-type line integral does not depend on the orientation of the curve. If $-\Gamma$ denotes the same curve with the opposite orientation, then

$$\boxed{\int_{-\Gamma} f\, ds = \int_\Gamma f\, ds.}$$

This follows from the fact that ds represents arc length and is always nonnegative.

3.4 Line Integrals of the Second Type

Let

$$\boldsymbol{F} = \boldsymbol{F}(x, y, z)$$

be a vector field defined in a region of $\mathbb{R}^3$, and let Γ be an oriented curve from A to B. Suppose a particle moves along Γ under the force $\boldsymbol{F}$. Divide Γ into small oriented displacements

$$\Delta \boldsymbol{r}_1, \ldots, \Delta \boldsymbol{r}_n,$$

and choose

$$\xi_i$$

on the i-th segment.
The work performed on that segment is approximately

$$\boldsymbol{F}(\xi_i) \cdot \Delta \boldsymbol{r}_i.$$

Thus the total work is approximated by

$$\sum_{i=1}^{n} \boldsymbol{F}(\xi_i) \cdot \Delta \boldsymbol{r}_i.$$

Definition 3.4.1. If

$$\lim_{\max\|\Delta \boldsymbol{r}_i\| \to 0} \sum_{i=1}^{n} \boldsymbol{F}(\xi_i) \cdot \Delta \boldsymbol{r}_i$$

exists and is independent of the division and the choice of sample points, then it is called the line integral of the second type of $\boldsymbol{F}$ along Γ, denoted by

$$\int_{\Gamma} \boldsymbol{F} \cdot d\boldsymbol{r}.$$

If

$$\boldsymbol{F} = (P, Q, R)$$

and

$$d\boldsymbol{r} = (dx, dy, dz),$$

then

$$\boldsymbol{F} \cdot d\boldsymbol{r} = P\,dx + Q\,dy + R\,dz.$$

Hence the second-type line integral is also written as

$$\boxed{\int_{\Gamma} P\,dx + Q\,dy + R\,dz.}$$

3.5 Computation of Second-Type Line Integrals

Suppose the oriented curve Γ is represented by

$$\boldsymbol{r}(t) = \Big(x(t), y(t), z(t)\Big), \qquad a \le t \le b,$$

where increasing t agrees with the prescribed orientation.
Then

$$d\boldsymbol{r} = \boldsymbol{r}'(t)\,dt.$$

Theorem 3.5.1. Under the corresponding assumptions,

$$\boxed{\int_\Gamma \boldsymbol{F}\cdot d\boldsymbol{r} = \int_a^b \boldsymbol{F}(\boldsymbol{r}(t))\cdot \boldsymbol{r}'(t)\,dt.}$$

If

$$\boldsymbol{F} = (P, Q, R),$$

then

$$\begin{aligned}\int_\Gamma P\,dx + Q\,dy + R\,dz = \int_a^b \Big[&P(x(t), y(t), z(t))x'(t)\\ &+ Q(x(t), y(t), z(t))y'(t)\\ &+ R(x(t), y(t), z(t))z'(t)\Big]\,dt.\end{aligned}$$

For a plane curve,

$$\boxed{\int_\Gamma P\,dx + Q\,dy = \int_a^b \Big(P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)\Big)\,dt.}$$

3.6 Orientation of Second-Type Line Integrals

The second-type line integral depends on the orientation.
If $-\Gamma$ denotes the same curve with the opposite orientation, then

$$\boxed{\int_{-\Gamma} \boldsymbol{F}\cdot d\boldsymbol{r} = -\int_\Gamma \boldsymbol{F}\cdot d\boldsymbol{r}.}$$

Similarly,

$$\int_{-\Gamma} P\,dx + Q\,dy + R\,dz = -\int_\Gamma P\,dx + Q\,dy + R\,dz.$$

3.7 Relation between the Two Types

Let

$$\boldsymbol{t}$$

be the unit tangent vector in the positive direction of Γ.
Since

$$d\boldsymbol{r} = \boldsymbol{t}\,ds,$$

we have

$$\boldsymbol{F}\cdot d\boldsymbol{r} = \boldsymbol{F}\cdot\boldsymbol{t}\,ds.$$

Hence

$$\boxed{\int_\Gamma \boldsymbol{F}\cdot d\boldsymbol{r} = \int_\Gamma \boldsymbol{F}\cdot\boldsymbol{t}\,ds.}$$

If

$$\boldsymbol{F} = (P, Q, R)$$

and

$$\boldsymbol{t} = \Big(\cos(\boldsymbol{t}, \boldsymbol{i}), \cos(\boldsymbol{t}, \boldsymbol{j}), \cos(\boldsymbol{t}, \boldsymbol{k})\Big),$$

then

$$\begin{aligned}\int_\Gamma P\,dx + Q\,dy + R\,dz = \int_\Gamma \Big[&P\cos(\boldsymbol{t}, \boldsymbol{i})\\ &+ Q\cos(\boldsymbol{t}, \boldsymbol{j})\\ &+ R\cos(\boldsymbol{t}, \boldsymbol{k})\Big]\,ds.\end{aligned}$$

3.8 Green's Formula

We now consider a plane region D with a piecewise smooth boundary

$$\partial D.$$

The positive orientation of ∂D is chosen so that, when a point moves along the boundary, the region D remains on its left.

Theorem 3.8.1 (Green's Formula)**.** Suppose $P(x,y)$ and $Q(x,y)$ have continuous first partial derivatives in a region containing D and its boundary. Then

$$\boxed{\oint_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.}$$

Sketch of Proof. First suppose that D is of the form

$$D = \{(x,y) : a \le x \le b,\ \varphi(x) \le y \le \psi(x)\}.$$

For the positively oriented boundary, the integrals over the vertical parts vanish in

$$\oint_{\partial D} P\,dx.$$

Thus

$$\begin{aligned}\oint_{\partial D} P\,dx = &\int_a^b P(x,\varphi(x))\,dx \\ &- \int_a^b P(x,\psi(x))\,dx.\end{aligned}$$

By the Newton–Leibniz formula,

$$P(x,\varphi(x)) - P(x,\psi(x)) = -\int_{\varphi(x)}^{\psi(x)} \frac{\partial P}{\partial y}\,dy.$$

Hence

$$\oint_{\partial D} P\,dx = -\iint_D \frac{\partial P}{\partial y}\,dx\,dy.$$

Similarly, for a region that can be described in the other direction,

$$\oint_{\partial D} Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dx\,dy.$$

Adding the two identities gives Green's formula. More general regions are obtained by decomposing them into finitely many such regions; integrals along common internal boundaries cancel. □

3.9 Area Formula from Green's Formula

Green's formula gives several expressions for the area of a plane region.
Taking

$$P = 0, \qquad Q = x,$$

we obtain

$$\boxed{\sigma(D) = \oint_{\partial D} x\,dy.}$$

Taking

$$P = -y, \qquad Q = 0,$$

we obtain

$$\boxed{\sigma(D) = -\oint_{\partial D} y\,dx.}$$

Combining these two formulas gives

$$\boxed{\sigma(D) = \frac{1}{2}\oint_{\partial D}\big(x\,dy - y\,dx\big).}$$

3.10 Green's Formula in Normal Form

Let

$$\boldsymbol{n}$$

be the outward unit normal vector to the positively oriented boundary ∂D.
If

$$\boldsymbol{n} = \Big(\cos(\boldsymbol{n},\boldsymbol{i}), \cos(\boldsymbol{n},\boldsymbol{j})\Big),$$

then the tangent and outward normal directions satisfy

$$\cos(\boldsymbol{t},\boldsymbol{i}) = -\cos(\boldsymbol{n},\boldsymbol{j}),$$

and

$$\cos(\boldsymbol{t},\boldsymbol{j}) = \cos(\boldsymbol{n},\boldsymbol{i}).$$

Green's formula can therefore be written in the form

$$\boxed{\int_{\partial D}\left[P\cos(\boldsymbol{n},\boldsymbol{i}) + Q\cos(\boldsymbol{n},\boldsymbol{j})\right]ds = \iint_D\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right)dx\,dy.}$$

3.11 Path Independence

Let $D \subset \mathbb{R}^2$ be a region, and consider

$$\int_\Gamma P\,dx + Q\,dy.$$

We say that the line integral is independent of path in D if its value depends only on the initial and terminal points of Γ.

Definition 3.11.1. A region D is called simply connected if every simple closed curve contained in D, together with the region enclosed by the curve, lies entirely in D.

Theorem 3.11.2. Let D be simply connected. Suppose P and Q have continuous first partial derivatives on D. Then the following conditions are equivalent:

1. for every closed curve Γ in D,

$$\oint_\Gamma P\,dx + Q\,dy = 0;$$

2. for any two curves Γ_1 and Γ_2 in D having the same initial and terminal points,

$$\int_{\Gamma_1} P\,dx + Q\,dy = \int_{\Gamma_2} P\,dx + Q\,dy;$$

3. there exists a function $u(x, y)$ on D such that

$$du = P\,dx + Q\,dy;$$

4.

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

throughout D.

Sketch of Proof. If the integral over every closed curve is zero, then joining two paths with the same endpoints in opposite directions gives a closed curve. This proves path independence.
Assume path independence. Fix a point

$$(a, b) \in D$$

and define

$$u(x, y) = \int_{(a,b)}^{(x,y)} P\,dx + Q\,dy.$$

Because the integral is independent of path, u is well defined.
Using a short horizontal segment,

$$u(x + h, y) - u(x, y) = \int_{x}^{x+h} P(t, y)\,dt.$$

Hence

$$u_x(x, y) = P(x, y).$$

Similarly,

$$u_y(x, y) = Q(x, y).$$

Therefore

$$du = P\,dx + Q\,dy.$$

If such a function u exists, then

$$P = u_x, \qquad Q = u_y.$$

Thus

$$\frac{\partial Q}{\partial x} = u_{yx} = u_{xy} = \frac{\partial P}{\partial y}.$$

Finally, if

$$Q_x = P_y,$$

Green's formula gives

$$\oint_{\Gamma} P\,dx + Q\,dy = 0$$

for every simple closed curve contained with its interior in D. The general closed-curve case follows by decomposition. □

3.12 Exact Differentials

If there exists a function $u(x, y)$ such that

$$du = P\,dx + Q\,dy,$$

then

$$P\,dx + Q\,dy$$

is called an exact differential.
In this case,

$$u_x = P, \qquad u_y = Q.$$

For any oriented curve Γ from

$$A = (x_1, y_1)$$

to

$$B = (x_2, y_2),$$

we have

$$\boxed{\int_\Gamma P\,dx + Q\,dy = u(x_2, y_2) - u(x_1, y_1).}$$

Thus the integral depends only on the two endpoints.

3.13 Multiply Connected Regions

The condition

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

does not by itself guarantee path independence on a multiply connected region.
For example, consider

$$P(x, y) = -\frac{y}{x^2 + y^2}, \qquad Q(x, y) = \frac{x}{x^2 + y^2}.$$

For

$$x^2 + y^2 > 0,$$

we have

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}.$$

However, if Γ is the positively oriented unit circle,

$$x = \cos t, \qquad y = \sin t, \qquad 0 \le t \le 2\pi,$$

then

$$\oint_\Gamma P\,dx + Q\,dy = 2\pi.$$

Thus the geometry of the region is essential in the criterion for path independence.

3.14 The Isoperimetric Problem

The isoperimetric problem asks:
Among all simple closed plane curves having a fixed length, which one encloses the greatest area?
The answer is the circle.

Theorem 3.14.1 (Isoperimetric Inequality)**.** Let Γ be a simple closed plane curve of length L, and let A be the area enclosed by Γ. Then

$$\boxed{A \leq \frac{L^2}{4\pi}.}$$

Equality holds for a circle.

Sketch of Proof. Parametrize Γ by arc length:

$$x = x(s), \qquad y = y(s), \qquad 0 \leq s \leq L.$$

Then

$$(x'(s))^2 + (y'(s))^2 = 1.$$

By the area formula derived from Green's theorem,

$$A = -\oint_\Gamma y\,dx = -\int_0^L y(s)x'(s)\,ds,$$

after choosing coordinates as in the proof.
The periodic functions involved satisfy the inequality used in the textbook, which gives

$$A \leq \frac{1}{2}\int_0^L \left[\left(\frac{L}{2\pi}x'(s)\right)^2 + y^2(s)\right] ds.$$

Together with

$$(x'(s))^2 + (y'(s))^2 = 1,$$

this yields

$$A \leq \frac{L^2}{4\pi}.$$

The equality conditions imply that the curve has the parametric form of a circle. Hence equality holds precisely for the circular case. □

Chapter 4

Surface Integrals

Line integrals integrate functions along curves. The corresponding construction on a surface leads to surface integrals.
As with line integrals, there are two types of surface integrals. The first type is defined using surface area and is independent of orientation. The second type is defined on oriented surfaces and depends on the chosen orientation.

4.1 Surface Area

Let a smooth surface Σ have a parametrization

$$\boldsymbol{r} = \boldsymbol{r}(u, v), \qquad (u, v) \in \Delta,$$

where

$$\boldsymbol{r}(u, v) = \Big(x(u, v), y(u, v), z(u, v)\Big).$$

Divide the parameter domain Δ into small regions

$$\Delta_1, \ldots, \Delta_n.$$

Each small region corresponds to a small piece of the surface. For sufficiently small Δ_i, the area of the corresponding surface piece is approximated by

$$\|\boldsymbol{r}_u \times \boldsymbol{r}_v\| \, \sigma(\Delta_i).$$

This leads to the definition of surface area.

Definition 4.1.1. Let

$$\boldsymbol{r} = \boldsymbol{r}(u, v), \qquad (u, v) \in \Delta,$$

be a regular parametrization of a surface Σ. Then the surface area of Σ is

$$\boxed{\sigma(\Sigma) = \iint_{\Delta} \|\boldsymbol{r}_u \times \boldsymbol{r}_v\| \; du \, dv.}$$

If

$$\boldsymbol{r}_u = (x_u, y_u, z_u)$$

and

$$\boldsymbol{r}_v = (x_v, y_v, z_v),$$

then

$$\|\boldsymbol{r}_u \times \boldsymbol{r}_v\| = \sqrt{EG - F^2},$$

where

$$E = x_u^2 + y_u^2 + z_u^2,$$

$$F = x_u x_v + y_u y_v + z_u z_v,$$

and

$$G = x_v^2 + y_v^2 + z_v^2.$$

Hence

$$\boxed{d\sigma = \sqrt{EG - F^2} \, du \, dv.}$$

4.2 Surface Given by a Graph

Suppose the surface is given by

$$z = f(x, y), \qquad (x, y) \in D.$$

A parametrization is

$$\boldsymbol{r}(x, y) = \Big(x, y, f(x, y)\Big).$$

Then

$$\boldsymbol{r}_x = (1, 0, f_x)$$

and

$$\boldsymbol{r}_y = (0, 1, f_y).$$

Therefore

$$\|\boldsymbol{r}_x \times \boldsymbol{r}_y\| = \sqrt{1 + f_x^2 + f_y^2}.$$

Thus

$$\boxed{d\sigma = \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\, dy.}$$

Consequently,

$$\boxed{\sigma(\Sigma) = \iint_D \sqrt{1 + f_x^2 + f_y^2}\, dx\, dy.}$$

4.3 Surface Integrals of the First Type

Suppose a mass is distributed over a surface Σ with surface density

$$\rho = \rho(x, y, z).$$

Divide Σ into small pieces

$$S_1, \ldots, S_n,$$

and choose

$$P_i \in S_i.$$

The mass of S_i is approximately

$$\rho(P_i)\sigma(S_i).$$

Thus the total mass is approximated by

$$\sum_{i=1}^{n} \rho(P_i)\sigma(S_i).$$

Definition 4.3.1. Let f be defined on a surface Σ. If

$$\lim_{\max \operatorname{diam}(S_i) \to 0} \sum_{i=1}^{n} f(P_i)\sigma(S_i)$$

exists and is independent of the division and the choice of P_i, then the limit is called the surface integral of the first type of f over Σ, denoted by

$$\int_\Sigma f\, d\sigma.$$

If $f \equiv 1$, then

$$\int_\Sigma d\sigma = \sigma(\Sigma).$$

The first-type surface integral does not depend on an orientation of the surface.

4.4 Computation of First-Type Surface Integrals

Suppose

$$\boldsymbol{r} = \boldsymbol{r}(u, v), \qquad (u, v) \in \Delta,$$

is a parametrization of Σ.
Then

$$\boxed{\int_\Sigma f \, d\sigma = \iint_\Delta f(\boldsymbol{r}(u, v)) \, \|\boldsymbol{r}_u \times \boldsymbol{r}_v\| \, du \, dv.}$$

If the surface is given by

$$z = \varphi(x, y), \qquad (x, y) \in D,$$

then

$$\boxed{\int_\Sigma f(x, y, z) \, d\sigma = \iint_D f(x, y, \varphi(x, y)) \sqrt{1 + \varphi_x^2 + \varphi_y^2} \, dx \, dy.}$$

4.5 Orientation of a Surface

To define the second type of surface integral, the surface must be oriented.
Let

$$\Sigma : \boldsymbol{r} = \boldsymbol{r}(u, v).$$

At each regular point,

$$\boldsymbol{r}_u \times \boldsymbol{r}_v$$

is perpendicular to the tangent plane.
Hence the two possible unit normal vectors are

$$\boldsymbol{n} = \pm \frac{\boldsymbol{r}_u \times \boldsymbol{r}_v}{\|\boldsymbol{r}_u \times \boldsymbol{r}_v\|}.$$

Definition 4.5.1. A surface is called orientable if one can choose a unit normal vector continuously over the entire surface.

Once one of the two continuous normal fields has been chosen, the surface is called an oriented surface.
A closed surface is usually oriented by its outward normal.
Not every surface is orientable. The Möbius strip is an example of a nonorientable surface.

4.6 Surface Integrals of the Second Type

Let Σ be an oriented surface and let

$$\boldsymbol{F} = (P, Q, R)$$

be a vector field defined on Σ.
Divide Σ into small oriented pieces S_i. Let $\boldsymbol{n}_i$ be the chosen unit normal vector at a point $P_i \in S_i$.
The quantity

$$\boldsymbol{F}(P_i) \cdot \boldsymbol{n}_i \, \sigma(S_i)$$

is the approximate flux through S_i.
This leads to the following definition.

Definition 4.6.1. The surface integral of the second type of $\boldsymbol{F}$ over Σ is

$$\boxed{\int_\Sigma \boldsymbol{F}\cdot\boldsymbol{n}\,d\sigma.}$$

If

$$\boldsymbol{n}=(\cos\alpha,\cos\beta,\cos\gamma),$$

then

$$\boldsymbol{F}\cdot\boldsymbol{n}=P\cos\alpha+Q\cos\beta+R\cos\gamma.$$

Thus

$$\boxed{\int_\Sigma \boldsymbol{F}\cdot\boldsymbol{n}\,d\sigma=\int_\Sigma\Big(P\cos\alpha+Q\cos\beta+R\cos\gamma\Big)d\sigma.}$$

Since

$$dy\,dz=\cos\alpha\,d\sigma,$$
$$dz\,dx=\cos\beta\,d\sigma,$$

and

$$dx\,dy=\cos\gamma\,d\sigma,$$

we may also write

$$\boxed{\int_\Sigma P\,dy\,dz+Q\,dz\,dx+R\,dx\,dy.}$$

4.7 Parametric Formula for the Second Type

Suppose

$$\Sigma:\boldsymbol{r}=\boldsymbol{r}(u,v),\qquad (u,v)\in\Delta.$$

Choose the orientation corresponding to

$$\boldsymbol{r}_u\times\boldsymbol{r}_v.$$

Then

$$\boldsymbol{n}\,d\sigma=\Big(\boldsymbol{r}_u\times\boldsymbol{r}_v\Big)\,du\,dv.$$

Therefore

$$\boxed{\int_\Sigma \boldsymbol{F}\cdot\boldsymbol{n}\,d\sigma=\iint_\Delta \boldsymbol{F}(\boldsymbol{r}(u,v))\cdot\Big(\boldsymbol{r}_u\times\boldsymbol{r}_v\Big)\,du\,dv.}$$

Equivalently,

$$\begin{aligned}&\int_\Sigma P\,dy\,dz+Q\,dz\,dx+R\,dx\,dy\\&\quad=\iint_\Delta\begin{vmatrix}P&Q&R\\x_u&y_u&z_u\\x_v&y_v&z_v\end{vmatrix}du\,dv.\end{aligned}$$

Changing the orientation changes the sign:

$$\int_{-\Sigma}\boldsymbol{F}\cdot\boldsymbol{n}\,d\sigma=-\int_\Sigma\boldsymbol{F}\cdot\boldsymbol{n}\,d\sigma.$$

4.8 Surface Given by a Graph

Suppose

$$\Sigma : z = f(x, y), \qquad (x, y) \in D.$$

For the orientation whose normal has positive z-component,

$$\boldsymbol{r}(x, y) = (x, y, f(x, y)),$$

and

$$\boldsymbol{r}_x \times \boldsymbol{r}_y = (-f_x, -f_y, 1).$$

Therefore

$$\boxed{\int_\Sigma P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy = \iint_D \big(-Pf_x - Qf_y + R\big)dx\,dy.}$$

For the opposite orientation, the sign is reversed.

4.9 Gauss Formula

Let $\Omega \subset \mathbb{R}^3$ be a bounded region whose boundary

$$\partial\Omega$$

is a piecewise smooth closed surface oriented by the outward normal.

Theorem 4.9.1 (Gauss Formula)**.** Suppose P, Q, R have continuous first partial derivatives in a region containing Ω and its boundary. Then

$$\boxed{\iint_{\partial\Omega} P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz.}$$

Equivalently, if

$$\boldsymbol{F} = (P, Q, R),$$

then

$$\boxed{\iint_{\partial\Omega} \boldsymbol{F} \cdot \boldsymbol{n}\,d\sigma = \iiint_\Omega (P_x + Q_y + R_z)\,dx\,dy\,dz.}$$

Sketch of Proof. Suppose first that

$$\Omega = \{(x, y, z) : (x, y) \in D,\ \varphi(x, y) \le z \le \psi(x, y)\}.$$

Consider the part involving $R\,dx\,dy$. The lateral surface has zero projection onto the xy-plane, while the upper and lower surfaces have opposite orientations. Therefore

$$\begin{aligned}\iint_{\partial\Omega} R\,dx\,dy = &\iint_D R(x, y, \psi(x, y))\,dx\,dy \\ &- \iint_D R(x, y, \varphi(x, y))\,dx\,dy.\end{aligned}$$

By the Newton–Leibniz formula,

$$\iint_{\partial\Omega} R\,dx\,dy = \iiint_\Omega \frac{\partial R}{\partial z}\,dx\,dy\,dz.$$

Similarly,

$$\iint_{\partial\Omega} P\,dy\,dz = \iiint_\Omega \frac{\partial P}{\partial x}\,dx\,dy\,dz,$$

and

$$\iint_{\partial\Omega} Q\,dz\,dx = \iiint_\Omega \frac{\partial Q}{\partial y}\,dx\,dy\,dz.$$

Adding the three formulas gives the result. □

4.10 Volume Formula from Gauss Formula

Taking

$$P = x, \qquad Q = y, \qquad R = z,$$

we have

$$P_x + Q_y + R_z = 3.$$

Hence Gauss' formula gives

$$\iint_{\partial\Omega} x\,dy\,dz + y\,dz\,dx + z\,dx\,dy = 3\mu(\Omega).$$

Therefore

$$\boxed{\mu(\Omega) = \frac{1}{3}\iint_{\partial\Omega} x\,dy\,dz + y\,dz\,dx + z\,dx\,dy.}$$

4.11 Stokes Formula

Let Σ be an oriented piecewise smooth surface whose boundary is the closed curve

$$\partial\Sigma.$$

The orientation of $\partial\Sigma$ and that of Σ are chosen consistently.

Theorem 4.11.1 (Stokes Formula)**.** Suppose P, Q, R have continuous first partial derivatives in a region containing Σ. Then

$$\boxed{\begin{aligned}\oint_{\partial\Sigma} P\,dx + Q\,dy + R\,dz = &\iint_{\Sigma}\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) dy\,dz \\ &+ \iint_{\Sigma}\left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right) dz\,dx \\ &+ \iint_{\Sigma}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.\end{aligned}}$$

Using the determinant notation in the textbook,

$$\boxed{\oint_{\partial\Sigma} P\,dx + Q\,dy + R\,dz = \iint_{\Sigma}\begin{vmatrix} dy\,dz & dz\,dx & dx\,dy \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.}$$

Equivalently, if

$$\boldsymbol{F} = (P, Q, R),$$

then

$$\oint_{\partial\Sigma} \boldsymbol{F}\cdot d\boldsymbol{r}$$

is expressed by the corresponding surface integral in the formula above.

4.12 Differential Forms

The formulas of Green, Gauss, and Stokes can be written in a common language using differential forms.
The exterior product satisfies

$$dx \wedge dx = 0,$$

$$dx \wedge dy = -dy \wedge dx,$$

and similarly,

$$dy \wedge dz = -dz \wedge dy,$$

$$dz \wedge dx = -dx \wedge dz.$$

For example,

$$dx \wedge dy \wedge dz = -dy \wedge dx \wedge dz.$$

4.13 Differential Forms in $\mathbb{R}^3$

A differential form of degree 0 is a function.
A differential form of degree 1 has the form

$$\omega = P\,dx + Q\,dy + R\,dz.$$

A differential form of degree 2 has the form

$$\omega = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy.$$

A differential form of degree 3 has the form

$$\omega = f\,dx \wedge dy \wedge dz.$$

4.14 Exterior Differentiation

For a function

$$f = f(x, y, z),$$

its exterior derivative is

$$\boxed{df = f_x\,dx + f_y\,dy + f_z\,dz.}$$

For the one-form

$$\omega = P\,dx + Q\,dy + R\,dz,$$

we define

$$d\omega = dP \wedge dx + dQ \wedge dy + dR \wedge dz.$$

Expanding and simplifying gives

$$\boxed{\begin{aligned} d\omega = {} & (R_y - Q_z)\,dy \wedge dz \\ & + (P_z - R_x)\,dz \wedge dx \\ & + (Q_x - P_y)\,dx \wedge dy. \end{aligned}}$$

For a two-form

$$\omega = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy,$$

we obtain

$$\boxed{d\omega = (P_x + Q_y + R_z)\,dx \wedge dy \wedge dz.}$$

4.15 Basic Properties of Exterior Differentiation

Exterior differentiation is linear:

$$d(\omega + \eta) = d\omega + d\eta.$$

For a constant c,

$$d(c\omega) = c\, d\omega.$$

A fundamental property is

$$\boxed{d(d\omega) = 0.}$$

In particular, for a function f,

$$d(df) = 0.$$

4.16 Green, Gauss, and Stokes in Differential Form

Let

$$\omega = P\, dx + Q\, dy.$$

Then

$$d\omega = (Q_x - P_y)\, dx \wedge dy.$$

Hence Green's formula can be written as

$$\boxed{\int_{\partial D} \omega = \int_D d\omega.}$$

For the two-form

$$\omega = P\, dy \wedge dz + Q\, dz \wedge dx + R\, dx \wedge dy,$$

we have

$$d\omega = (P_x + Q_y + R_z)\, dx \wedge dy \wedge dz.$$

Thus Gauss' formula also has the form

$$\boxed{\int_{\partial \Omega} \omega = \int_\Omega d\omega.}$$

Similarly, Stokes' formula can be written as

$$\boxed{\int_{\partial \Sigma} \omega = \int_\Sigma d\omega.}$$

Therefore Green's formula, Gauss' formula, and Stokes' formula are different instances of the same relation:

$$\boxed{\int_{\partial M} \omega = \int_M d\omega.}$$

Chapter 5

Applications of Integration

The integral was introduced as an analytic construction, but many of its most important uses come from geometry and physics. The basic idea is to divide a quantity into small pieces, approximate each piece, add the approximations, and then pass to a limit.
This chapter collects several applications of this principle. We first consider areas, arc lengths, volumes, and surfaces of revolution. We then give several physical examples, use geometric area comparisons to estimate sums, and conclude with the Wallis and Stirling formulas.

5.1 Applications of Integration in Geometry

The geometric applications of integration are based on the same general procedure:

$$\text{partition} \longrightarrow \text{approximation} \longrightarrow \text{summation} \longrightarrow \text{limit}.$$

5.2 Areas of Plane Figures

Suppose

$$f(x) \geq 0, \qquad a \leq x \leq b.$$

The area of the region bounded by

$$x = a, \qquad x = b, \qquad y = 0, \qquad y = f(x)$$

is

$$\boxed{S = \int_a^b f(x)\,dx.}$$

More generally, suppose f and g are continuous on $[a, b]$ and

$$f(x) \geq g(x), \qquad a \leq x \leq b.$$

Then the area between the two curves

$$y = f(x) \qquad \text{and} \qquad y = g(x)$$

is

$$\boxed{S = \int_a^b \Big(f(x) - g(x)\Big)\,dx.}$$

Similarly, if

$$\varphi(y) \geq \psi(y), \qquad c \leq y \leq d,$$

then the area between

$$x = \varphi(y) \qquad \text{and} \qquad x = \psi(y)$$

is

$$\boxed{S = \int_c^d \Big(\varphi(y) - \psi(y)\Big)\,dy.}$$

If a plane curve is given in polar coordinates by

$$r = r(\theta), \qquad \alpha \leq \theta \leq \beta,$$

then the area swept out by the radius vector is

$$\boxed{S = \frac{1}{2}\int_\alpha^\beta r^2(\theta)\,d\theta.}$$

Example 5.2.1. For the cardioid

$$r = a(1 + \cos\theta), \qquad a > 0,$$

the enclosed area is

$$\begin{aligned} S &= 2 \cdot \frac{1}{2} \int_0^{\pi} a^2 (1 + \cos\theta)^2 \, d\theta \\ &= a^2 \int_0^{\pi} \Big(1 + 2\cos\theta + \cos^2\theta\Big) \, d\theta \\ &= \frac{3}{2} \pi a^2. \end{aligned}$$

5.3 Arc Length of a Space Curve

Let a C^1 space curve Γ be represented by

$$\boldsymbol{r} = \boldsymbol{r}(t), \qquad \alpha \le t \le \beta,$$

where

$$\boldsymbol{r}(t) = \Big(x(t), y(t), z(t)\Big).$$

Divide

$$[\alpha, \beta]$$

by

$$\alpha = t_0 < t_1 < \cdots < t_n = \beta.$$

The corresponding points on the curve are

$$A_i = \boldsymbol{r}(t_i), \qquad i = 0, 1, \ldots, n.$$

The polygonal approximation to the arc length is

$$\sum_{i=1}^{n} |A_i - A_{i-1}| \, .$$

Passing to the limit gives the arc-length formula.

Theorem 5.3.1. Let

$$\Gamma : \boldsymbol{r} = \boldsymbol{r}(t), \qquad \alpha \le t \le \beta,$$

be a C^1 curve. Then

$$\boxed{s(\Gamma) = \int_{\alpha}^{\beta} \|\boldsymbol{r}'(t)\| \, dt.}$$

In coordinates,

$$\boxed{s(\Gamma) = \int_{\alpha}^{\beta} \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} \, dt.}$$

For a plane curve

$$x = x(t), \qquad y = y(t),$$

we have

$$s(\Gamma) = \int_{\alpha}^{\beta} \sqrt{(x'(t))^2 + (y'(t))^2} \, dt.$$

If

$$y = f(x), \qquad a \le x \le b,$$

then

$$\boxed{s(\Gamma) = \int_a^b \sqrt{1 + (f'(x))^2}\, dx.}$$

If the curve is represented in polar coordinates by

$$r = r(\theta), \qquad \alpha \le \theta \le \beta,$$

then

$$\boxed{s(\Gamma) = \int_\alpha^\beta \sqrt{r^2(\theta) + (r'(\theta))^2}\, d\theta.}$$

Example 5.3.2. Consider the astroid

$$x = a\cos^3 t, \qquad y = a\sin^3 t, \qquad 0 \le t \le 2\pi.$$

By symmetry,

$$\begin{aligned} s(\Gamma) &= 4\int_0^{\pi/2} \sqrt{(x'(t))^2 + (y'(t))^2}\, dt \\ &= 12a \int_0^{\pi/2} \sin t \cos t\, dt \\ &= 6a. \end{aligned}$$

5.4 Volumes of Spatial Regions

Let Ω be a spatial region lying between the planes

$$x = a \qquad \text{and} \qquad x = b.$$

Suppose the cross-sectional area perpendicular to the x-axis is

$$g(x).$$

Divide $[a, b]$ into small intervals. The volume of a thin slice is approximately

$$g(\xi_k)\Delta x_k.$$

Passing to the limit gives

$$\boxed{V = \int_a^b g(x)\, dx.}$$

In particular, suppose

$$y = f(x) \ge 0, \qquad a \le x \le b,$$

and rotate the region below the curve about the x-axis.
The cross-sectional area is

$$g(x) = \pi f^2(x).$$

Hence the volume of the solid of revolution is

$$\boxed{V = \pi \int_a^b f^2(x)\, dx.}$$

Example 5.4.1. The volume of a sphere of radius a is obtained by rotating

$$y = \sqrt{a^2 - x^2}, \qquad -a \le x \le a,$$

about the x-axis.
Thus

$$\begin{aligned} V &= \pi \int_{-a}^{a} (a^2 - x^2)\, dx \\ &= 2\pi \int_{0}^{a} (a^2 - x^2)\, dx \\ &= \frac{4}{3}\pi a^3. \end{aligned}$$

5.5 Areas of Surfaces of Revolution

Let

$$\Gamma : \qquad x = x(t), \qquad y = y(t), \qquad \alpha \le t \le \beta,$$

be a C^1 plane curve lying in the upper half-plane.
Rotate Γ about the x-axis. The resulting surface has area

$$\boxed{S = 2\pi \int_{\alpha}^{\beta} y(t)\sqrt{(x'(t))^2 + (y'(t))^2}\, dt.}$$

Since

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2}\, dt,$$

the formula may also be written as

$$S = 2\pi \int_{\Gamma} y\, ds.$$

If

$$y = f(x) \ge 0, \qquad a \le x \le b,$$

then

$$\boxed{S = 2\pi \int_{a}^{b} f(x)\sqrt{1 + (f'(x))^2}\, dx.}$$

5.6 Examples of Physical Applications

Integration is also used to accumulate continuously distributed physical quantities.
One example concerns gravitational attraction.
Suppose a uniform rod AB has length l and mass M. A particle C of mass m lies on the extension of the rod, with

$$CA = a.$$

Choose the origin at C. A small piece of the rod of length dx has mass

$$\frac{M}{l}\, dx.$$

Its gravitational force on C is

$$dF = G\frac{m(M/l)\, dx}{x^2}.$$

Therefore

$$\begin{aligned} F &= \frac{GmM}{l} \int_{a}^{a+l} \frac{dx}{x^2} \\ &= \boxed{\frac{GmM}{a(a+l)}}. \end{aligned}$$

5.7 Work in a Gravitational Field

Let a body of mass m be at distance x from the center of the Earth, whose mass is M. The gravitational force has magnitude

$$F(x) = G\frac{mM}{x^2}.$$

The work required to move the body from distance h to infinity is

$$\begin{aligned} W_h &= \int_h^{+\infty} F(x)\,dx \\ &= GmM\int_h^{+\infty} \frac{dx}{x^2} \\ &= \boxed{\frac{GmM}{h}}. \end{aligned}$$

5.8 Escape Velocity

Suppose a body of mass m is launched vertically from the surface of the Earth.
Let R be the radius of the Earth, M its mass, and v_0 the initial velocity required for the body to reach infinity.
The initial kinetic energy must equal the work needed to overcome the gravitational attraction:

$$\frac{1}{2}mv_0^2 = \frac{GmM}{R}.$$

Since

$$mg = G\frac{mM}{R^2},$$

we have

$$GM = gR^2.$$

Therefore

$$\boxed{v_0 = \sqrt{2gR}.}$$

Using the numerical values in the textbook,

$$v_0 \approx 11.17 \text{ km/s}.$$

This is called the escape velocity.

5.9 The Area Principle

Many estimates involving finite sums can be obtained by comparing the sum with the area under a curve.
This method will be referred to as the area principle.

Theorem 5.9.1. Let

$$x \geq m, \qquad m \in \mathbb{N}^+,$$

and suppose f is a nonnegative increasing function. Then

$$\boxed{\left|\sum_{k=m}^{[\xi]} f(k) - \int_m^{\xi} f(x)\,dx\right| \leq f(\xi).}$$

Proof. For every integer k,

$$f(k-1) \le \int_{k-1}^{k} f(x)\,dx \le f(k).$$

Summing from $k = m+1$ to n, we obtain

$$\begin{aligned} f(m) + f(m+1) + \cdots + f(n-1) &\le \int_m^n f(x)\,dx \\ &\le f(m+1) + \cdots + f(n). \end{aligned}$$

Combining this inequality with the remaining interval from $[\xi]$ to ξ yields the stated estimate. □

For example, if

$$f(x) = x^\lambda, \qquad \lambda > 0,$$

then

$$\boxed{\sum_{n=m}^{[\xi]} n^\lambda = \frac{\xi^{\lambda+1} - m^{\lambda+1}}{\lambda+1} + O(\xi^\lambda).}$$

5.10 Area Principle for Decreasing Functions

Theorem 5.10.1. Let

$$x \ge m, \qquad m \in \mathbb{N}^+,$$

and suppose f is a nonnegative decreasing function. Then the limit

$$\boxed{\lim_{n\to\infty} \left(\sum_{k=m}^{n} f(k) - \int_m^n f(x)\,dx \right) = \alpha}$$

exists, where

$$0 \le \alpha \le f(m).$$

If, in addition,

$$\lim_{x\to+\infty} f(x) = 0,$$

then

$$\boxed{\left| \sum_{k=m}^{[\xi]} f(k) - \int_m^{\xi} f(x)\,dx - \alpha \right| \le f(\xi - 1).}$$

Sketch of Proof. Define

$$g(n) = \sum_{k=m}^{n} f(k) - \int_m^n f(x)\,dx.$$

Since f is decreasing,

$$g(n) - g(n+1) = -f(n+1) + \int_n^{n+1} f(x)\,dx \ge 0.$$

Hence $\{g(n)\}$ is decreasing. Moreover,

$$g(n) \ge 0.$$

Thus $g(n)$ has a finite limit α.
The error estimate follows by comparing the remaining integral with the corresponding endpoint values of f. □

5.11 Euler's Constant

Take

$$f(x) = \frac{1}{x}, \qquad m = 1.$$

Then

$$\lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \ln n \right) = \gamma,$$

where γ is Euler's constant.
Moreover,

$$\boxed{\sum_{k=1}^{n} \frac{1}{k} = \ln n + \gamma + O\left(\frac{1}{n}\right).}$$

5.12 An Estimate for the Alternating Harmonic Series

Consider

$$\sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k}.$$

Let

$$S_n = \sum_{k=1}^{n} (-1)^{k-1} \frac{1}{k}.$$

Then

$$\begin{aligned} S_{2n} &= \left(1 + \frac{1}{3} + \cdots + \frac{1}{2n-1}\right) \\ &\quad - \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}\right). \end{aligned}$$

Using the asymptotic formula for the harmonic sum,

$$\begin{aligned} S_{2n} &= \left(1 + \frac{1}{2} + \cdots + \frac{1}{2n}\right) \\ &\quad - 2\left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}\right) \\ &= \ln 2 + O\left(\frac{1}{n}\right). \end{aligned}$$

Therefore

$$\boxed{\sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} = \ln 2.}$$

5.13 A Logarithmic Inequality

Let

$$0 < a < b.$$

By comparing the area under the curve

$$y = \frac{1}{x}$$

with suitable trapezoids and rectangles, we obtain

$$\boxed{\frac{2}{a+b} < \frac{\ln b - \ln a}{b-a} < \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right).}$$

In particular, taking

$$a = n, \qquad b = n + 1,$$

gives

$$\boxed{\frac{1}{n+1/2} < \ln\left(1 + \frac{1}{n}\right) < \frac{1}{2}\left(\frac{1}{n} + \frac{1}{n+1}\right).}$$

This inequality will be used in the derivation of Stirling's formula.

5.14 Young's Inequality

Let

$$\varphi : [0, +\infty) \to [0, +\infty)$$

be continuous and strictly increasing, with

$$\varphi(0) = 0.$$

Then φ has an inverse function

$$\varphi^{-1}.$$

Theorem 5.14.1 (Young's Inequality)**.** For $a, b > 0$,

$$\boxed{ab \leq \int_0^a \varphi(x)\,dx + \int_0^b \varphi^{-1}(y)\,dy.}$$

Equality holds if and only if

$$b = \varphi(a).$$

Take

$$\varphi(x) = x^{p-1}, \qquad p > 1.$$

Then

$$\varphi^{-1}(y) = y^{q-1},$$

where

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Young's inequality becomes

$$\boxed{ab \leq \frac{a^p}{p} + \frac{b^q}{q}.}$$

Equality holds if and only if

$$a^p = b^q.$$

5.15 Hölder's Inequality

Theorem 5.15.1 (Hölder's Inequality)**.** Let

$$a_1, \ldots, a_n, \qquad b_1, \ldots, b_n$$

be nonnegative real numbers. Let $p, q > 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$\boxed{\sum_{i=1}^n a_i b_i \leq \left(\sum_{i=1}^n a_i^p\right)^{1/p} \left(\sum_{i=1}^n b_i^q\right)^{1/q}.}$$

Proof. Set

$$A = \sum_{i=1}^{n} a_i^p, \qquad B = \sum_{i=1}^{n} b_i^q.$$

Apply Young's inequality to

$$\frac{a_i}{A^{1/p}} \qquad \text{and} \qquad \frac{b_i}{B^{1/q}}.$$

Then

$$\frac{a_i b_i}{A^{1/p} B^{1/q}} \le \frac{1}{p}\frac{a_i^p}{A} + \frac{1}{q}\frac{b_i^q}{B}.$$

Summing over i gives

$$\frac{\sum_{i=1}^{n} a_i b_i}{A^{1/p} B^{1/q}} \le \frac{1}{p} + \frac{1}{q} = 1.$$

Therefore

$$\sum_{i=1}^{n} a_i b_i \le A^{1/p} B^{1/q}.$$

□

When

$$p = q = 2,$$

Hölder's inequality becomes the Cauchy–Schwarz inequality:

$$\boxed{\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).}$$

There is also an integral form:

$$\boxed{\int_a^b |f(x)g(x)|\,dx \le \left(\int_a^b |f(x)|^p\,dx\right)^{1/p} \left(\int_a^b |g(x)|^q\,dx\right)^{1/q}.}$$

5.16 Wallis Formula

For

$$0 < x < \frac{\pi}{2},$$

we have

$$0 < \sin x < 1.$$

Hence

$$\sin^{2n+1} x < \sin^{2n} x < \sin^{2n-1} x.$$

Integrating from 0 to $\pi/2$,

$$\int_0^{\pi/2} \sin^{2n+1} x\,dx < \int_0^{\pi/2} \sin^{2n} x\,dx < \int_0^{\pi/2} \sin^{2n-1} x\,dx.$$

Using the reduction formulas for these integrals gives

$$\frac{(2n)!!}{(2n+1)!!} < \frac{(2n-1)!!}{(2n)!!}\frac{\pi}{2} < \frac{(2n-2)!!}{(2n-1)!!}.$$

After rearrangement,

$$\frac{2n}{2n+1}\frac{\pi}{2} < \frac{((2n)!!)^2}{(2n+1)((2n-1)!!)^2} < \frac{\pi}{2}.$$

By the squeeze theorem,

$$\boxed{\lim_{n\to\infty} \frac{1}{2n+1}\left(\frac{2\cdot 4\cdots 2n}{1\cdot 3\cdots(2n-1)}\right)^2 = \frac{\pi}{2}.}$$

This is Wallis' formula.
An equivalent form is

$$\boxed{\sqrt{\pi} = \lim_{n\to\infty} \frac{(n!)^2 2^{2n}}{(2n)!\sqrt{n}}.}$$

5.17 Stirling Formula

Stirling's formula gives a simple asymptotic approximation for $n!$.

Theorem 5.17.1 (Stirling Formula)**.** As $n \to \infty$,

$$\boxed{n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.}$$

Sketch of Proof. Define

$$a_n = \frac{n!e^n}{n^{n+1/2}}.$$

Then

$$\frac{a_n}{a_{n+1}} = \frac{1}{e}\left(1+\frac{1}{n}\right)^{n+1/2}.$$

Taking logarithms,

$$\ln\frac{a_n}{a_{n+1}} = \left(n+\frac{1}{2}\right)\ln\left(1+\frac{1}{n}\right) - 1.$$

Using the logarithmic inequality established above,

$$0 < \left(n+\frac{1}{2}\right)\ln\left(1+\frac{1}{n}\right) - 1 < \frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right).$$

Hence

$$1 < \frac{a_n}{a_{n+1}} < \exp\left[\frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right)\right].$$

Thus $\{a_n\}$ is decreasing and bounded below, so

$$a_n \to \alpha$$

for some $\alpha > 0$.
Using Wallis' formula,

$$\alpha^2 = \lim_{n\to\infty} \frac{(n!)^2 e^{2n}}{n^{2n+1}} = 2\pi.$$

Therefore

$$\alpha = \sqrt{2\pi}.$$

Hence

$$\frac{n!e^n}{n^{n+1/2}} \to \sqrt{2\pi},$$

which is equivalent to

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

□

Chapter 6

Point-Set Topology in Euclidean Space

6.1 n-Dimensional Euclidean Space

The Cartesian product of n copies of $\mathbb{R}$ is denoted by

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{R},\ i = 1, 2, \ldots, n\}.$$

An element

$$x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$$

is called a *point* or a *vector* in $\mathbb{R}^n$.
For

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n),$$

define

$$x + y = (x_1 + y_1, \ldots, x_n + y_n)$$

and, for $\lambda \in \mathbb{R}$,

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

The zero vector is

$$0 = (0, \ldots, 0),$$

and

$$-x = (-x_1, \ldots, -x_n).$$

Definition 6.1.1 (Inner Product)**.** For $x, y \in \mathbb{R}^n$, define

$$(x, y) = \sum_{i=1}^{n} x_i y_i.$$

This is called the *inner product* of x and y.

The inner product satisfies

$$(x, y) = (y, x),$$
$$(x + y, z) = (x, z) + (y, z),$$
$$(\lambda x, y) = \lambda (x, y),$$

and

$$(x, x) \geq 0,$$

where equality holds if and only if $x = 0$.

Definition 6.1.2 (Norm)**.** For $x \in \mathbb{R}^n$, define

$$\|x\| = \sqrt{(x, x)} = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}.$$

The number $\|x\|$ is called the *norm* of x.

Theorem 6.1.3 (Cauchy–Schwarz Inequality)**.** For any $x, y \in \mathbb{R}^n$,

$$|(x, y)| \leq \|x\| \, \|y\|.$$

Proof. If $y = 0$, the result is immediate. Suppose $y \neq 0$. For every $t \in \mathbb{R}$,

$$0 \leq \|x - ty\|^2 = \|x\|^2 - 2t(x, y) + t^2\|y\|^2.$$

The discriminant of this quadratic polynomial in t is nonpositive. Hence

$$4(x, y)^2 - 4\|x\|^2\|y\|^2 \leq 0,$$

which gives

$$|(x, y)| \leq \|x\| \, \|y\|.$$

□

Corollary 6.1.4 (Triangle Inequality)**.** For any $x, y \in \mathbb{R}^n$,

$$\|x + y\| \leq \|x\| + \|y\|.$$

Proof. By the Cauchy–Schwarz inequality,

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + 2(x, y) + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2 . \end{aligned}$$

Taking square roots proves the result. □

For $x, y \in \mathbb{R}^n$, the distance between x and y is defined by

$$d(x, y) = \|x - y\|.$$

Thus

$$d(x, y) \geq 0, \qquad d(x, y) = 0 \iff x = y,$$
$$d(x, y) = d(y, x),$$

and

$$d(x, z) \leq d(x, y) + d(y, z).$$

Definition 6.1.5 (Open Ball)**.** For $a \in \mathbb{R}^n$ and $r > 0$, the set

$$B_r(a) = \{x \in \mathbb{R}^n : \|x - a\| < r\}$$

is called the *open ball* with center a and radius r.

Similarly,

$$\overline{B}_r(a) = \{x \in \mathbb{R}^n : \|x - a\| \leq r\}$$

is called the *closed ball* with center a and radius r.

6.2 Limits of Point Sequences in $\mathbb{R}^n$

Let

$$x^{(k)} = \left(x_1^{(k)}, x_2^{(k)}, \ldots, x_n^{(k)}\right) \in \mathbb{R}^n, \qquad k \in \mathbb{N}.$$

Then

$$\left\{x^{(k)}\right\}_{k=1}^{\infty}$$

is called a *point sequence* in $\mathbb{R}^n$.

Definition 6.2.1 (Limit of a Point Sequence)**.** Let $a \in \mathbb{R}^n$. We say that

$$x^{(k)} \to a \qquad (k \to \infty)$$

if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$k > N \quad \Longrightarrow \quad \|x^{(k)} - a\| < \varepsilon.$$

In this case,

$$\lim_{k\to\infty} x^{(k)} = a.$$

Equivalently, every neighborhood of a contains all but finitely many terms of the sequence.

Theorem 6.2.2 (Coordinatewise Characterization)**.** Let

$$x^{(k)} = \left(x_1^{(k)}, \ldots, x_n^{(k)}\right), \qquad a = (a_1, \ldots, a_n).$$

Then

$$x^{(k)} \to a$$

if and only if

$$x_i^{(k)} \to a_i, \qquad i = 1, \ldots, n.$$

Proof. If $x^{(k)} \to a$, then

$$\left|x_i^{(k)} - a_i\right| \le \|x^{(k)} - a\|,$$

so each coordinate converges.
Conversely, suppose

$$x_i^{(k)} \to a_i \qquad (i = 1, \ldots, n).$$

For any $\varepsilon > 0$, choose N such that, for $k > N$,

$$\left|x_i^{(k)} - a_i\right| < \frac{\varepsilon}{\sqrt{n}} \qquad (i = 1, \ldots, n).$$

Then

$$\|x^{(k)} - a\|^2 = \sum_{i=1}^{n} \left|x_i^{(k)} - a_i\right|^2 < \varepsilon^2.$$

Hence $x^{(k)} \to a$. □

Definition 6.2.3 (Cauchy Sequence)**.** A point sequence $\{x^{(k)}\} \subset \mathbb{R}^n$ is called a *Cauchy sequence* if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$k, l > N \quad \Longrightarrow \quad \|x^{(k)} - x^{(l)}\| < \varepsilon.$$

Theorem 6.2.4 (Cauchy Criterion)**.** A point sequence in $\mathbb{R}^n$ converges if and only if it is a Cauchy sequence.

Sketch of Proof. By the coordinatewise characterization, $\{x^{(k)}\}$ is Cauchy in $\mathbb{R}^n$ if and only if each coordinate sequence

$$\left\{x_i^{(k)}\right\}_{k=1}^{\infty}$$

is Cauchy in $\mathbb{R}$. The result therefore follows from the completeness of $\mathbb{R}$. □

Theorem 6.2.5 (Bolzano–Weierstrass Theorem)**.** Every bounded point sequence in $\mathbb{R}^n$ has a convergent subsequence.

Sketch of Proof. Let

$$x^{(k)} = \left(x_1^{(k)}, \ldots, x_n^{(k)}\right)$$

be bounded. Then each coordinate sequence $\{x_i^{(k)}\}$ is bounded.
First choose a subsequence on which the first coordinate converges. From this subsequence choose another subsequence on which the second coordinate converges. Continue this process through all n coordinates.
The resulting subsequence has every coordinate convergent, and hence converges in $\mathbb{R}^n$. □

6.3 Open Sets and Closed Sets in $\mathbb{R}^n$

Definition 6.3.1 (Interior Point)**.** Let $E \subset \mathbb{R}^n$. A point $a \in E$ is called an *interior point* of E if there exists $r > 0$ such that

$$B_r(a) \subset E.$$

The set of all interior points of E is called the *interior* of E, denoted by

$$E^{\circ}.$$

Definition 6.3.2 (Open Set)**.** A set $E \subset \mathbb{R}^n$ is called *open* if every point of E is an interior point of E, that is,

$$E = E^{\circ}.$$

Theorem 6.3.3. Every open ball $B_r(a)$ is an open set.

Proof. Let $x \in B_r(a)$. Then

$$\|x - a\| < r.$$

Set

$$\rho = r - \|x - a\| > 0.$$

If $y \in B_\rho(x)$, then

$$\|y - a\| \leq \|y - x\| + \|x - a\| < \rho + \|x - a\| = r.$$

Thus $B_\rho(x) \subset B_r(a)$. □

Theorem 6.3.4 (Properties of Open Sets)**.** The following statements hold:

1. $\varnothing$ and $\mathbb{R}^n$ are open;
2. the union of any family of open sets is open;

3. the intersection of finitely many open sets is open.

Definition 6.3.5 (Neighborhood)**.** If $a \in \mathbb{R}^n$, any open set containing a is called a *neighborhood* of a.

Definition 6.3.6 (Limit Point)**.** Let $E \subset \mathbb{R}^n$. A point $a \in \mathbb{R}^n$ is called a *limit point* of E if every neighborhood of a contains a point of E different from a.

Equivalently, a is a limit point of E if

$$(B_r(a) \setminus \{a\}) \cap E \neq \varnothing$$

for every $r > 0$.

Theorem 6.3.7 (Sequential Characterization of Limit Points)**.** A point a is a limit point of E if and only if there exists a sequence

$$x^{(k)} \in E \setminus \{a\}$$

such that

$$x^{(k)} \to a.$$

Proof. Suppose a is a limit point of E. For each $k \in \mathbb{N}$, choose

$$x^{(k)} \in \left(B_{1/k}(a) \setminus \{a\}\right) \cap E.$$

Then

$$0 < \|x^{(k)} - a\| < \frac{1}{k},$$

so $x^{(k)} \to a$.
Conversely, if

$$x^{(k)} \in E \setminus \{a\}, \qquad x^{(k)} \to a,$$

then every neighborhood of a contains $x^{(k)}$ for all sufficiently large k. Hence a is a limit point of E. □

Definition 6.3.8 (Closed Set)**.** A set $E \subset \mathbb{R}^n$ is called *closed* if it contains all of its limit points.

Theorem 6.3.9. A set $E \subset \mathbb{R}^n$ is closed if and only if its complement

$$E^c = \mathbb{R}^n \setminus E$$

is open.

Theorem 6.3.10 (Properties of Closed Sets)**.** The following statements hold:

1. $\varnothing$ and $\mathbb{R}^n$ are closed;
2. the intersection of any family of closed sets is closed;
3. the union of finitely many closed sets is closed.

Definition 6.3.11 (Closure)**.** Let $E \subset \mathbb{R}^n$. The union of E and all of its limit points is called the *closure* of E, denoted by

$$\overline{E}.$$

Equivalently,

$$a \in \overline{E}$$

if and only if

$$B_r(a) \cap E \neq \varnothing$$

for every $r > 0$.

Theorem 6.3.12. For $E \subset \mathbb{R}^n$, $\overline{E}$ is the smallest closed set containing E.

Definition 6.3.13 (Boundary Point)**.** A point $a \in \mathbb{R}^n$ is called a *boundary point* of E if every neighborhood of a contains both a point of E and a point of E^c.

The set of all boundary points of E is denoted by

$$\partial E.$$

We have

$$\partial E = \overline{E} \cap \overline{E^c}.$$

6.4 Sequentially Compact Sets and Compact Sets

Definition 6.4.1 (Sequentially Compact Set)**.** A set $E \subset \mathbb{R}^n$ is called *sequentially compact* if every sequence in E has a subsequence converging to a point of E.

Theorem 6.4.2. A subset $E \subset \mathbb{R}^n$ is sequentially compact if and only if it is bounded and closed.

Proof. Suppose first that E is sequentially compact.
If E were unbounded, for each $k \in \mathbb{N}$ we could choose $x^{(k)} \in E$ such that

$$\|x^{(k)}\| > k.$$

This sequence cannot have a convergent subsequence, a contradiction. Thus E is bounded. Now let a be a limit point of E. There exists

$$x^{(k)} \in E \setminus \{a\}$$

such that $x^{(k)} \to a$. Since E is sequentially compact, some subsequence converges to a point of E. But every subsequence of $x^{(k)}$ converges to a, so $a \in E$. Hence E is closed.
Conversely, suppose E is bounded and closed. Every sequence in E is bounded, so by the Bolzano–Weierstrass theorem it has a convergent subsequence. Since E is closed, the limit belongs to E. Hence E is sequentially compact. □

Definition 6.4.3 (Open Cover)**.** Let $E \subset \mathbb{R}^n$. A family of open sets $\{G_\alpha\}_{\alpha \in A}$ is called an *open cover* of E if

$$E \subset \bigcup_{\alpha \in A} G_\alpha.$$

Definition 6.4.4 (Compact Set)**.** A set $E \subset \mathbb{R}^n$ is called *compact* if every open cover of E contains a finite subcover.

Theorem 6.4.5 (Heine–Borel Theorem)**.** For $E \subset \mathbb{R}^n$, the following conditions are equivalent:

1. E is compact;

2. E is sequentially compact;

3. E is closed and bounded.

Sketch of Proof. The equivalence between sequential compactness and closedness plus boundedness has already been proved.
It remains to relate sequential compactness and compactness. If an open cover has no finite subcover, repeatedly subdividing a bounded rectangle containing E produces a nested sequence of closed rectangles whose diameters tend to zero and each of which cannot be covered by finitely many members of the original cover. Their common limiting point belongs to E, and hence lies in one member of the open cover. A sufficiently small rectangle must then lie in that member, giving a contradiction.
Thus a closed and bounded subset of $\mathbb{R}^n$ is compact. The converse follows by showing that an unbounded set, or a set missing one of its limit points, admits an open cover with no finite subcover. □

Corollary 6.4.6. Every closed subset of a compact set is compact.

6.5 Connected Sets

Definition 6.5.1 (Separated Sets)**.** Two subsets $A, B \subset \mathbb{R}^n$ are called *separated* if

$$A \cap \overline{B} = \varnothing$$

and

$$\overline{A} \cap B = \varnothing.$$

Definition 6.5.2 (Connected Set)**.** A set $E \subset \mathbb{R}^n$ is called *connected* if it cannot be represented as

$$E = A \cup B,$$

where A and B are nonempty separated sets.

Equivalently, E is connected if it cannot be written as the union of two nonempty disjoint sets which are both relatively open in E.

Theorem 6.5.3. An interval $I \subset \mathbb{R}$ is connected.

Proof. Suppose

$$I = A \cup B, \qquad A \neq \varnothing, \qquad B \neq \varnothing,$$

where A and B are separated.
Choose $a \in A$ and $b \in B$, with $a < b$. Since I is an interval,

$$[a, b] \subset I.$$

Let

$$c = \sup\left(A \cap [a, b]\right).$$

Using the defining property of the supremum, one obtains a sequence of points of A converging to c. Hence

$$c \in \overline{A}.$$

A similar argument using points immediately to the right of c shows that $c \in \overline{B}$, contradicting the separation of A and B. □

Definition 6.5.4 (Path)**.** A continuous mapping

$$\varphi : [a, b] \to \mathbb{R}^n$$

is called a *path*.

If

$$\varphi(a) = p, \qquad \varphi(b) = q,$$

then the path is said to join p and q.

Definition 6.5.5 (Path-Connected Set)**.** A set $E \subset \mathbb{R}^n$ is called *path-connected* if any two points of E can be joined by a path lying entirely in E.

Theorem 6.5.6. Every path-connected set is connected.

Proof. Suppose E is path-connected but not connected. Then

$$E = A \cup B$$

for two nonempty separated sets A and B.
Choose $p \in A$ and $q \in B$. Since E is path-connected, there exists a continuous mapping

$$\varphi : [a, b] \to E$$

such that

$$\varphi(a) = p, \qquad \varphi(b) = q.$$

The interval $[a, b]$ is connected, and the continuous image of a connected set is connected. Hence $\varphi([a, b])$ is connected, which is impossible because it meets both A and B. □

Definition 6.5.7 (Region)**.** A nonempty open connected subset of $\mathbb{R}^n$ is called a *region*.

Theorem 6.5.8. Every region in $\mathbb{R}^n$ is path-connected.

Sketch of Proof. Fix $a \in D$, where D is a region, and let A be the set of all points in D that can be joined to a by a polygonal path lying in D.
Since D is open, sufficiently small balls around points of A remain in D; hence A is relatively open in D. The same argument shows that $D \setminus A$ is relatively open. Since D is connected and $A \neq \varnothing$, we must have

$$A = D.$$

Thus any point of D can be joined to a, and hence any two points of D can be joined by a polygonal path. □

6.6 Limits of Multivariable Functions

Let

$$D \subset \mathbb{R}^n, \qquad f : D \to \mathbb{R},$$

and suppose a is a limit point of D.

Definition 6.6.1 (Limit of a Multivariable Function)**.** We say that

$$\lim_{x \to a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in D, \qquad 0 < \|x - a\| < \delta$$

imply

$$|f(x) - L| < \varepsilon.$$

For a function

$$f(x, y),$$

the notation

$$\lim_{(x,y) \to (a,b)} f(x, y) = L$$

means that $f(x, y)$ approaches L whenever (x, y) approaches (a, b) in an arbitrary manner. This is essentially different from approaching the point along only one prescribed curve.

Remark 6.6.2. To prove that

$$\lim_{(x,y) \to (a,b)} f(x, y)$$

does not exist, it is enough to find two different paths approaching (a, b) along which $f(x, y)$ has different limiting values.

Theorem 6.6.3 (Sequential Criterion)**.** Let a be a limit point of D. Then

$$\lim_{x \to a} f(x) = L$$

if and only if for every sequence

$$x^{(k)} \in D \setminus \{a\}, \qquad x^{(k)} \to a,$$

we have

$$f\left(x^{(k)}\right) \to L.$$

Proof. Suppose first that

$$\lim_{x \to a} f(x) = L.$$

For every $\varepsilon > 0$, choose $\delta > 0$ such that

$$0 < \|x - a\| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Since $x^{(k)} \to a$, the inequality

$$0 < \|x^{(k)} - a\| < \delta$$

holds for all sufficiently large k. Hence

$$f\left(x^{(k)}\right) \to L.$$

Conversely, suppose the limit is not L. Then there exists $\varepsilon_0 > 0$ such that for every $k \in \mathbb{N}$, one can choose

$$x^{(k)} \in D$$

with

$$0 < \|x^{(k)} - a\| < \frac{1}{k},$$

but

$$\left|f\left(x^{(k)}\right) - L\right| \geq \varepsilon_0.$$

Then $x^{(k)} \to a$, while $f(x^{(k)}) \not\to L$, a contradiction. □

Theorem 6.6.4 (Algebra of Limits)**.** Suppose

$$\lim_{x\to a} f(x) = L, \qquad \lim_{x\to a} g(x) = M.$$

Then

$$\lim_{x\to a}[f(x) + g(x)] = L + M,$$

$$\lim_{x\to a} f(x)g(x) = LM,$$

and, if $M \neq 0$,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Theorem 6.6.5 (Cauchy Criterion)**.** Let

$$f : D \to \mathbb{R},$$

and let a be a limit point of D. The finite limit

$$\lim_{x\to a} f(x)$$

exists if and only if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x', x'' \in D, \qquad 0 < \|x' - a\| < \delta, \qquad 0 < \|x'' - a\| < \delta$$

imply

$$|f(x') - f(x'')| < \varepsilon.$$

Sketch of Proof. Necessity follows immediately from

$$|f(x') - f(x'')| \leq |f(x') - L| + |f(x'') - L|.$$

For sufficiency, choose any sequence

$$x^{(k)} \to a, \qquad x^{(k)} \neq a.$$

The assumed condition implies that

$$\left\{f\left(x^{(k)}\right)\right\}$$

is a Cauchy sequence in $\mathbb{R}$, hence convergent. The condition also forces all such sequences to have the same limit. The sequential criterion then gives the existence of $\lim_{x\to a} f(x)$. □

6.7 Continuous Functions of Several Variables

Definition 6.7.1 (Continuity at a Point)**.** Let

$$f : D \subset \mathbb{R}^n \to \mathbb{R}, \qquad a \in D.$$

The function f is said to be *continuous at* a if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in D, \qquad \|x - a\| < \delta$$

imply

$$|f(x) - f(a)| < \varepsilon.$$

If a is a limit point of D, this is equivalent to

$$\lim_{x \to a} f(x) = f(a).$$

Definition 6.7.2 (Continuity on a Set)**.** A function $f : D \to \mathbb{R}$ is called *continuous on* D if it is continuous at every point of D.

Theorem 6.7.3 (Sequential Criterion for Continuity)**.** The function $f : D \to \mathbb{R}$ is continuous at $a \in D$ if and only if

$$x^{(k)} \to a, \qquad x^{(k)} \in D$$

implies

$$f\left(x^{(k)}\right) \to f(a).$$

Theorem 6.7.4. If f and g are continuous at a, then

$$f + g, \qquad fg$$

are continuous at a. If $g(a) \neq 0$, then

$$\frac{f}{g}$$

is also continuous at a.

Theorem 6.7.5 (Extreme Value Theorem)**.** Let $D \subset \mathbb{R}^n$ be compact and let

$$f : D \to \mathbb{R}$$

be continuous. Then f is bounded on D, and there exist $x_m, x_M \in D$ such that

$$f(x_m) = \min_{x \in D} f(x),$$

and

$$f(x_M) = \max_{x \in D} f(x).$$

Proof. Suppose f is unbounded. Then there exists a sequence $\{x^{(k)}\} \subset D$ such that

$$|f(x^{(k)})| > k.$$

Since D is compact, a subsequence satisfies

$$x^{(k_j)} \to a \in D.$$

By continuity,

$$f(x^{(k_j)}) \to f(a),$$

contradicting

$$|f(x^{(k_j)})| \to \infty.$$

Thus f is bounded.
Let

$$M = \sup_{x \in D} f(x).$$

Choose $x^{(k)} \in D$ such that

$$M - \frac{1}{k} < f(x^{(k)}) \leq M.$$

Compactness gives a subsequence

$$x^{(k_j)} \to x_M \in D.$$

By continuity,

$$f(x_M) = \lim_{j \to \infty} f(x^{(k_j)}) = M.$$

The minimum is obtained similarly. □

Theorem 6.7.6 (Intermediate Value Theorem)**.** Let $D \subset \mathbb{R}^n$ be connected and let

$$f : D \to \mathbb{R}$$

be continuous. If $a, b \in D$ and

$$f(a) < c < f(b),$$

then there exists $x \in D$ such that

$$f(x) = c.$$

Proof. The image $f(D) \subset \mathbb{R}$ of a connected set under a continuous function is connected. Hence $f(D)$ is an interval. Since

$$f(a), f(b) \in f(D),$$

every number between $f(a)$ and $f(b)$ also belongs to $f(D)$. Thus there exists $x \in D$ such that $f(x) = c$. □

Definition 6.7.7 (Uniform Continuity)**.** Let $D \subset \mathbb{R}^n$. A function

$$f : D \to \mathbb{R}$$

is called *uniformly continuous on* D if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in D$,

$$\|x - y\| < \delta$$

implies

$$|f(x) - f(y)| < \varepsilon.$$

Theorem 6.7.8 (Heine–Cantor Theorem)**.** Every continuous function on a compact subset of $\mathbb{R}^n$ is uniformly continuous.

Proof. Suppose $f : D \to \mathbb{R}$ is continuous on compact D, but is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ and sequences

$$x^{(k)}, y^{(k)} \in D$$

such that

$$\|x^{(k)} - y^{(k)}\| < \frac{1}{k},$$

while

$$\left|f(x^{(k)}) - f(y^{(k)})\right| \geq \varepsilon_0.$$

By compactness, some subsequence satisfies

$$x^{(k_j)} \to a \in D.$$

Since

$$\|x^{(k_j)} - y^{(k_j)}\| \to 0,$$

we also have

$$y^{(k_j)} \to a.$$

Continuity gives

$$f(x^{(k_j)}) \to f(a), \qquad f(y^{(k_j)}) \to f(a),$$

so

$$\left|f(x^{(k_j)}) - f(y^{(k_j)})\right| \to 0,$$

a contradiction. □

6.8 Continuous Mappings

Let

$$D \subset \mathbb{R}^n.$$

A mapping

$$f : D \to \mathbb{R}^m$$

can be written as

$$f(x) = (f_1(x), f_2(x), \ldots, f_m(x)),$$

where

$$f_i : D \to \mathbb{R}, \qquad i = 1, \ldots, m.$$

Definition 6.8.1 (Limit of a Vector-Valued Mapping)**.** Let a be a limit point of D, and let

$$p \in \mathbb{R}^m.$$

We write

$$\lim_{x \to a} f(x) = p$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in D, \qquad 0 < \|x - a\| < \delta$$

imply

$$\|f(x) - p\| < \varepsilon.$$

Theorem 6.8.2 (Coordinatewise Characterization)**.** Let

$$f = (f_1, \ldots, f_m), \qquad p = (p_1, \ldots, p_m).$$

Then

$$\lim_{x \to a} f(x) = p$$

if and only if

$$\lim_{x \to a} f_i(x) = p_i, \qquad i = 1, \ldots, m.$$

Definition 6.8.3 (Continuous Mapping)**.** A mapping

$$f : D \subset \mathbb{R}^n \to \mathbb{R}^m$$

is called *continuous at* $a \in D$ if

$$\lim_{x \to a} f(x) = f(a),$$

whenever a is a limit point of D.
The mapping is called *continuous on* D if it is continuous at every point of D.

Theorem 6.8.4. The mapping

$$f = (f_1, \ldots, f_m)$$

is continuous at a if and only if every component function f_i is continuous at a.

Theorem 6.8.5 (Composition of Continuous Mappings)**.** Let

$$f : D \subset \mathbb{R}^n \to \mathbb{R}^m$$

be continuous at a, and let

$$g : E \subset \mathbb{R}^m \to \mathbb{R}^k$$

be continuous at $f(a)$, where $f(D) \subset E$. Then

$$g \circ f$$

is continuous at a.

Theorem 6.8.6 (Inverse Image Characterization)**.** A mapping

$$f : D \to \mathbb{R}^m$$

is continuous on D if and only if, for every open set $G \subset \mathbb{R}^m$,

$$f^{-1}(G)$$

is relatively open in D.

Proof. Suppose f is continuous and let $G \subset \mathbb{R}^m$ be open. If

$$x \in f^{-1}(G),$$

then $f(x) \in G$. Since G is open, there exists $\varepsilon > 0$ such that

$$B_\varepsilon(f(x)) \subset G.$$

By continuity, there exists $\delta > 0$ such that

$$y \in D, \qquad \|y - x\| < \delta$$

implies

$$\|f(y) - f(x)\| < \varepsilon.$$

Hence

$$B_\delta(x) \cap D \subset f^{-1}(G),$$

so $f^{-1}(G)$ is relatively open in D.
Conversely, suppose the inverse image of every open set is relatively open in D. For $x \in D$ and $\varepsilon > 0$,

$$B_\varepsilon(f(x))$$

is open. Therefore

$$f^{-1}\left(B_\varepsilon(f(x))\right)$$

is a relatively open neighborhood of x. Hence there exists $\delta > 0$ such that

$$B_\delta(x) \cap D \subset f^{-1}\left(B_\varepsilon(f(x))\right).$$

Thus

$$\|y - x\| < \delta \quad \Longrightarrow \quad \|f(y) - f(x)\| < \varepsilon,$$

which proves continuity at x. □

Theorem 6.8.7. The continuous image of a compact set is compact.

Proof. Let $D \subset \mathbb{R}^n$ be compact and let

$$f : D \to \mathbb{R}^m$$

be continuous.
Let

$$\{G_\alpha\}_{\alpha \in A}$$

be an open cover of $f(D)$. Then

$$\left\{f^{-1}(G_\alpha)\right\}_{\alpha \in A}$$

is an open cover of D. Since D is compact, there exist $\alpha_1, \ldots, \alpha_N$ such that

$$D \subset \bigcup_{j=1}^{N} f^{-1}(G_{\alpha_j}).$$

Therefore

$$f(D) \subset \bigcup_{j=1}^{N} G_{\alpha_j}.$$

Hence $f(D)$ is compact. □

Theorem 6.8.8. The continuous image of a connected set is connected.

Proof. Let D be connected and suppose $f(D)$ is not connected. Then there exist nonempty separated subsets $A, B \subset f(D)$ such that

$$f(D) = A \cup B.$$

The inverse images

$$f^{-1}(A), \qquad f^{-1}(B)$$

give a separation of D, contradicting the connectedness of D. Therefore $f(D)$ is connected.
□

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He also makes extensive use of examples and counterexamples. These examples do not simply demonstrate how a theorem is applied; they help clarify why each hypothesis is needed, where a conclusion may fail, and how related concepts should be distinguished. This style of teaching gradually taught me to pay attention not only to mathematical results, but also to the logical structure behind them.
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